The Structural Frictions of Industrial Decarbonization

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Abstract

Restricting grants to existing plants avoids paying new plants that would be built anyway, but existing plants may need much larger grants to adopt a cleaner technology. To study this trade-off, I estimate a dynamic structural model of the U.S. steel industry in which old plants keep running, close or adopt the cleaner technology, beside their old one or in its place, and new plants enter. Estimated on a new plant-level panel, it includes two structural frictions that old plants face and new ones do not: obligations to workers due at shutdown, and a fixed adoption cost, estimated from choices, that exceeds the cost of the new equipment. At equal abatement, this restriction cuts the share paid for investment that would happen anyway from about half to an eighth but barely changes public spending per tonne of CO_2 avoided, because old plants need larger grants. Paying only plants that replace their old technology nearly doubles that spending: owners would rather add the cleaner technology beside it and defer the obligations.

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1 Introduction

A clean technology can spread in two ways: existing plants adopt it, or new plants built around it enter and compete with them. Supporting cleaner production can therefore preserve existing industrial sites or shift production toward new ones, with different consequences for workers, owners and taxpayers. In 2024 the U.S. Department of Energy selected projects to reduce industrial emissions at existing and new plants alike, for up to USD 6 billion in grants (DOE 2024a), while its grants for converting auto plants were only for existing facilities, to retain workers (DOE 2024b). This raises a question: do clean-investment grants restricted to existing plants cost governments less per tonne of CO_2 avoided than grants open to new plants too? Open grants have low additionality when they pay for investment that would happen without support, but excluding new plants can rule out abatement at lower public cost.

To study this trade-off, I estimate a dynamic structural model of the U.S. steel industry. Old plants, called integrated works, make steel from iron ore and coal, using blast furnaces, while the cleaner electric arc furnace (EAF) melts scrap with electricity. Each year these old works decide whether to keep running, close, or adopt the EAF, beside their blast furnaces, which keep running (an *add-on*), or in their place (a *conversion*), and potential new EAF mills decide whether to enter. The model captures both sides of the trade-off. New mills can enter without a grant, so an open grant pays for some entry that would happen anyway, and their output lowers steel prices, which pushes old works to close. An EAF at an old works takes over steel that its blast furnaces would have made, part of it in an add-on and all of it in a conversion, and with it the profit that steel earned, which a new mill does not give up. The old works also bears two *structural frictions* that a new mill does not: obligations to its workers due when the blast furnaces close, and a fixed adoption cost, so it may need a much larger grant. Because owners look ahead, and entry moves the prices that old works respond to, the policy experiments solve adoption, closure, entry and steel prices together, for every plant and year.

I ask what it costs governments to decarbonize an industry through its old plants rather than through new ones, and why. Three questions make this concrete. Does restricting clean-investment grants to existing plants lower the public cost of abatement? What does adopting the cleaner technology cost an old plant beyond the new equipment? And what decides whether an old plant adds the new technology beside the old one or replaces it? I study them in steel, but the same questions arise in cement, chemicals and other heavy industries, where the same grant programs fund both existing and new plants.

The paper makes three contributions. First, I show that higher additionality, the share of payments that changes investment, need not lower the public cost of abatement. The recipients a rule excludes because they would invest anyway may also be those whose added investment costs least per tonne of CO_2 avoided, and the larger payments needed elsewhere can then offset the saving. Second, I show from owners' choices that adopting a cleaner technology costs an old plant more than the new equipment and

the profit its old capital gives up. This helps explain why old plants need much larger grants than new ones, and why restricting grants to them saves so little. Third, I show that obligations to workers shape not only when old plants close but also how they adopt: because they fall due when the old capital is retired, they steer old plants toward adding the new technology beside the old one rather than replacing it. This helps make a grant paid only for replacement the most expensive.

Steelmaking accounts for 7 to 8 percent of human-caused greenhouse gas emissions (World Steel Association 2025), and in the United States the shift to the cleaner technology has already happened. The EAF's share of production rose from 34% in 1985 to 71% in 2020, new mills account for 75% of the fall in the CO_2 intensity of steelmaking capacity, and only three old works adopted the EAF. Because these choices were made without grants of the kind studied here, they show which investment comes without them. I build a year-by-year panel of U.S. steel plants for 1985–2020, mostly from the industry's plant directories, which I digitized. It records each plant's furnaces with their capacities, openings, closures and idle years, and its owners with the dates they changed, and I match each owner to the unfunded pension and retiree-health obligations it reported in financial data and company filings. The panel lets me observe each plant's choices, every adoption, closure and entry, together with the obligations the owner carried at the time.

Each year, the owner of a plant chooses the action with the highest expected value, given current profits, the cost of each action and an unobserved shock to each choice, and owners forecast steel margins from their past movements. Profits come from observed prices and engineering input requirements, and the building cost of an EAF from industry cost estimates. I estimate five parameters, among them the fixed adoption cost and the scale of the unobserved shocks, from owners' yearly choices by maximum likelihood with a nested fixed point (Rust 1987), and then the costs of setting up new mills from the observed entry. The fixed adoption cost is estimated from how rarely old works adopted, given what adopting would have earned, which varied across works and years with steel margins, the works' size and the owner's obligations. They passed it up in many years, which points to a large cost, and the two adoptions in the estimation period limit how large it can be. How large the fixed cost is in dollars depends on how strongly owners' choices respond to profits. EAF mills, which do not face this cost, help pin this down: how often they close as their margins rise and fall shows how strongly their choices respond.

In the policy experiments, I compare three clean-investment grants per tonne of new EAF capacity, each set to avoid the same CO_2 over 1985–2015: an open grant for all plants, an adoption grant for existing plants, and a conversion grant for existing plants that replace their old technology. Restricting grants to existing plants sharply raises additionality but barely changes public spending per tonne of CO_2 avoided. Under the open grant, 54 percent of the payments go to investment that would happen anyway, mostly to new mills, against 13 percent under the adoption grant. Yet spending per tonne avoided is about the same, 27.5 against 28.1 USD. Counting only the payments

for investment the grant causes, the adoption grant spends almost twice as much per tonne as the open grant, because old works need much larger grants to adopt, and the open grant's lower additionality closes the difference. Restricting saves more when little abatement is bought. The ranking reverses if the costs of setting up new mills are at the high end of their estimated range or steel demand responds weakly to prices, and at the upper end of the fixed adoption cost's interval the adoption grant cannot avoid that much CO_2. Restricting grants keeps more jobs at existing plants, while the open grant creates more at new mills, leaving similar net job losses.

The estimates show why old works need much larger grants. I estimate the fixed adoption cost at 0.77 billion USD per adoption, about 414 USD per tonne of new EAF capacity, against a building cost of 260 USD per tonne. The fixed adoption cost may include stranded ore mines and coke batteries, the time customers take to approve the new steel, and permitting. I split the *adoption gap*, the one-time payment that would just make an owner willing to add an EAF, into its parts. Net of the value of closing, about a quarter of the gap is the new furnace itself, its building cost and the products it can make, much as for a new mill. The rest is specific to old works: a fifth is the profit the blast furnaces give up, and about half the structural frictions, almost all of it the fixed adoption cost.

Paying only for conversions makes almost every payment additional, yet nearly doubles spending, to 49.5 USD per tonne, and costs the most at every level of abatement I study, because owners prefer an add-on. The fixed adoption cost discourages both forms of adoption, but the obligations add 134 USD per tonne of capacity to the gap to convert and little to the gap to an add-on, which keeps the blast furnaces running. A conversion also leaves the works without steel for the years the new furnace takes to build, and with a smaller furnace, so a grant must be much larger to make owners convert.

Related literature. First, the paper contributes to the literature on the targeting of green and industrial subsidies. Much of this literature concerns additionality: Boomhower and Davis (2014), Xing, Leard, and Li (2021) and Calel et al. (2025) find that rebates for efficient appliances, credits for electric vehicles and carbon offsets mostly pay recipients who would have adopted anyway, and Sallee (2025) ties the revenue cost of green subsidies to that share. I contribute by showing that higher additionality need not lower the public cost of abatement when adoption by existing plants competes with entry. The closest paper is Barwick, Kalouptsidi, and Zahur (2025), who find that entry subsidies were the least effective support for Chinese shipbuilders because they drew in high-cost firms. However, in my setting the entrants are new mills that need much smaller grants than old works, which is why excluding them saves so little.

Second, the paper contributes to the literature on technology diffusion through incumbents and entrants. In steel and cement, new technologies spread both through adoption by existing plants (Oster 1982; Macher, Miller, and Osborne 2021) and through the entry of new plants and the closure of old ones (Collard-Wexler and De Loecker

2015). Explanations of late adoption by incumbents point to sunk investment in the old technology (Hornbeck et al. 2026) and output lost while switching over (Holmes, Levine, and Schmitz 2012), and costs owed to workers can delay closure (Deily 1987). The closest paper is Igami (2017): the cannibalization of their existing products made incumbent hard-disk makers reluctant to introduce a new generation, and an incumbent that adopts keeps making the old generation beside the new. I contribute by making that a choice: old works can add the new technology or replace the old. I also show from owners' choices that the estimated fixed cost of adopting exceeds both the new furnace and the profit the old capital gives up, the counterpart of cannibalization.

Third, the paper contributes to the literature on climate and environmental policy for heavy industry, and in particular on how it treats existing and new plants. Much of this literature studies carbon prices and their design, from border carbon adjustments with capacity held fixed (Clausing et al. 2025; Park, Rausch, and Karplus 2026) to the adoption of abatement technologies as a carbon price rises (Gorrens 2025). How such rules treat existing and new plants changes who produces: Ryan (2012) finds that regulation that raised the cost of entry in U.S. cement benefited incumbents, and Pacelli (2026) that free permits, granted only to operating plants, act as entry subsidies that draw in dirtier entrants. Fowlie, Reguant, and Ryan (2016) compare auctioned permits with free permits for existing capacity, which entrants do not receive, in a dynamic model of U.S. cement in which entrants bring the frontier technology. I contribute by showing that obligations to workers, which fall due when old capital is retired, shape how existing plants respond to such policy: they steer old works toward adding the cleaner technology rather than replacing the old, which helps make a grant paid only for replacement the most expensive.

Section 2 describes the two ways of making steel, the plants, their owners and the panel. Section 3 sets out the facts the model must explain, Section 4 presents the model and the adoption gap, and Section 5 the estimates. Section 6 compares the grants with each other and with a carbon price at equal abatement, reports their welfare and who gains and who pays, and asks how large a grant the works still running blast furnaces would need to add an EAF fed with iron made from natural gas. Section 7 concludes.

2 Background and Data

2.1 Production routes and adoption

An integrated works makes iron from ore. Coke, made from coal in batteries on or near the site, reduces iron ore to liquid iron in a blast furnace, and a basic oxygen furnace (BOF) blows oxygen through the iron, with some scrap, to make steel. In what follows, a works' blast furnaces stand for its blast furnaces and the oxygen furnaces they feed. A blast furnace runs for a campaign of many years and must then be relined, which means rebuilding its worn heat-resistant inner lining, or shut. A works is large and feeds its own rolling and finishing mills. I use 2.0 tonnes of CO_2 per tonne of steel made this way, the rounded cradle-to-gate estimate for a modeled typical U.S. plant in Zang et al.

(2023). In the model, the need to reline arrives as a chance event, after which the owner chooses whether to reline or take another available action (Section 4.1).

An electric arc furnace (EAF) melts steel scrap with electricity. A mill built around one needs no blast furnace, coke or ore and can be small; its emissions come mostly from the power plants that supply it, and the counterfactuals use 0.4 tonnes of CO_2 per tonne of EAF steel. It is also cheaper to build. In 2005 dollars, an EAF steelmaking shop, the furnace and the casters that solidify its steel, costs 260 USD per tonne of capacity to replace, and a whole EAF mill with its rolling mills 586 USD. I use the replacement cost of an EAF steelmaking shop as the building cost of adoption (Section 4.2).

EAF mills reached the higher-value products in steps. Steel is sold as long products (bars, rebar, wire rod and shapes) or flat products (sheet and plate), and among sheet, cold-rolled, coated and tin-plated grades sell for more than hot-rolled. EAF mills first made only long products. From 1989, a new casting method, thin slabs rolled directly into sheet, let them make hot-rolled sheet (Peters 1991); they later added cold rolling and coating. Scrap carries copper and tin that cannot be refined out and limit the best sheet, so mills making it mix in iron made from ore, which costs more. The model takes these technological improvements as given rather than allowing plants or policy to determine their timing (Section 4.1).

An old works can adopt the EAF in two ways. In an *add-on*, it builds an EAF beside its blast furnaces, which can keep running, and becomes a hybrid that may retire them later, or never. In a *conversion*, it shuts the blast furnaces and builds an EAF in their place. Both reuse the works' casters and rolling mills, and I call both *adoption*. I reserve *replacement* for the other margin: a new EAF mill, built elsewhere and usually by another firm, takes over the market of works that close. In 1985, 5 works already ran an EAF beside their blast furnaces. The model lets an integrated works choose either form of adoption and lets a hybrid retire its blast furnaces later (Section 4.1).

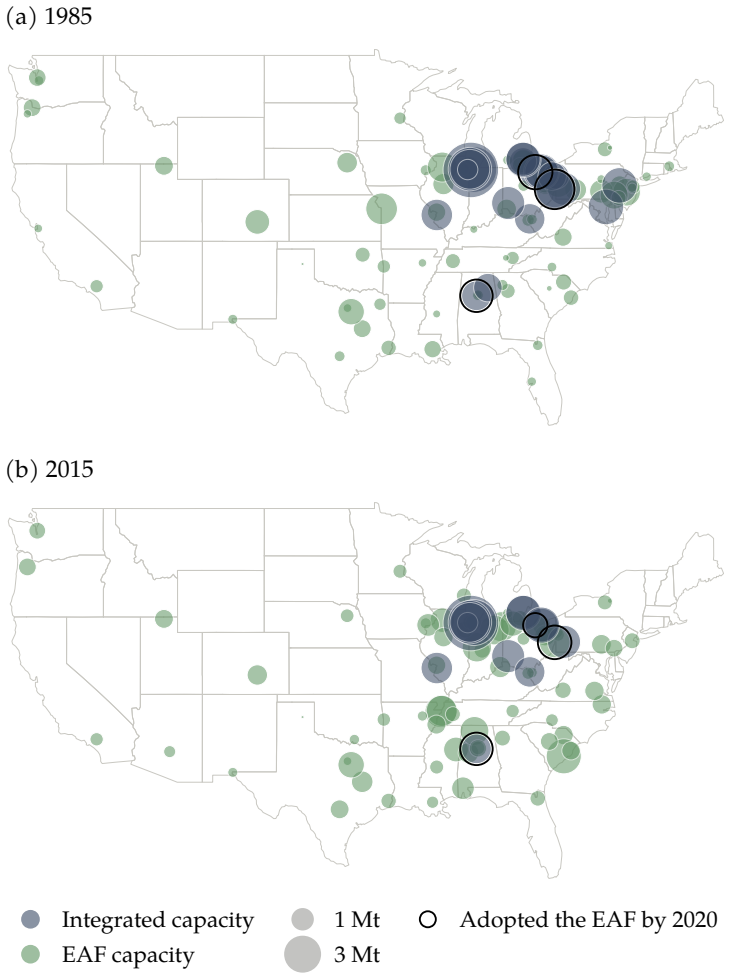
2.2 Plants, owners and shutdown costs

Across the 166 sites of the panel, integrated steelmaking capacity (BOF capacity, including that of hybrids) fell from 72.7 million tonnes in 1985 to 41.6 in 2020, and EAF capacity rose from 64.6 to 80.0. EAF capacity is nameplate and ran at lower utilization than integrated capacity, so its share exceeds the EAF share of steel production. The old works sat around the Great Lakes, Pittsburgh and Ohio; the new mills spread across the country, most of all into the South (Giarratani, Madhavan, and Gruver 2013), in different places from the capacity that closed (Figure 1).

The integrated firms owed their workers and retirees pensions beyond what their plans held, and retiree health care. I call this stock the *obligations*. When a works closes, part of it falls due, with extra shutdown benefits; these payments go to workers and retirees. Cleaning up the site also falls due at closure and, unlike the payments, uses real resources. I call the payments and the cleanup together the *shutdown costs*.

The sums due at closure could be large. When the federal pension insurer sued to terminate the plan of Wheeling-Pittsburgh's parent, it argued that contingent plant-

Figure 1: U.S. carbon-steel plants in 1985 and 2015



Notes: Carbon-steel sites in the estimation scope (126) in 1985 and 2015. Circle area is capacity, split into the integrated part (blast furnaces with basic oxygen furnaces) and the EAF part, so a works with both shows two circles. Ringed: the three works that adopted the EAF by 2020. EAF: electric arc furnace.

shutdown liabilities could exceed USD 378 million. At Republic, shutdown benefits could add about USD 95 million to plans already less than half funded.[1] When an owner went bankrupt, the Pension Benefit Guaranty Corporation (PBGC), the federal insurer of private pension plans, could take over its pension plans, and their obligations were then no longer the owner's. The PBGC does not take over retiree health care. In the model, closing or converting triggers payments to workers and site-cleanup costs, while adding an EAF does not trigger them at construction (Section 4.2).

2.3 The plant panel

Each edition of the plant directories of the Association for Iron & Steel Technology (AIST 2004–2022) and its predecessor, the Association of Iron and Steel Engineers (AISE 1983–2003), lists the steelmaking shops of U.S. plants with their owners and equipment. I digitized most editions from library copies and extracted their entries into one record

[1] Sources: WHX Corporation (2003), 10-K for fiscal 2002; United States Court of Appeals, Sixth Circuit (2004), opinion in the Republic Technologies bankruptcy.

per site and year: 6,401 site-years from 1984 to 2021 (Appendix A.1). It gives each site's owner, with changes dated by completed deals because the directories report them late (Appendix A.3), and coke batteries and finishing lines.

I reconcile the directory record with the steel plant capacity reports of King (2025), a proprietary database that lists iron- and steelmaking units by plant, with each unit's opening, modification and closing years and its capacity. I check discrepancies against company filings, environmental reports and industry equipment surveys, including the roundups of the Iron & Steel Society (ISS 1988–2003), published in *Iron & Steelmaker*. These surveys help check early furnace capacities. The reconciliation yields 118 dated corrections to the King-based record; 91 cite the directories or the equipment surveys. I change a King entry only where the evidence establishes a correction; otherwise it stands, and its continuous unit histories fill the years between directory editions. Every site in the estimation sample is confirmed by at least one source besides the King reports. The result is a panel of 4,384 site-years at 166 sites, 1985–2020. I treat its capacities as nameplate capacity, in million metric tonnes, and mark a year with no production while the capacity stays on the books as idle. The model takes plant capacities from the panel and fixes the capacity changes associated with adoption rather than letting owners choose their size (Section 4.1).

A firm's obligations are its pension plans' projected benefit obligation less plan assets, plus its retiree health obligation, from Compustat (S&P Global Market Intelligence 2026b) and the notes to 10-K filings. Cleanup costs come from the environmental accruals in the integrated firms' annual reports from 1993 on; firms book only amounts they judge probable, so these accruals are lower bounds (Appendix A.3). The model ties each works' shutdown charge to its owner (Section 4.2).

The model concerns carbon-steel plants, which could replace an integrated works or be replaced by one; I leave out 40 specialty-steel, stainless, foundry and forge shops, which sell in other markets (Appendix A.2). That leaves 126 sites: 26 integrated works and 100 EAF mills. Of these, 122 operate in at least one year of 1985–2014, which gives 2,763 decision years, each with its outcome read from the following year (through 2015); the decisions of 2015 to 2019 are held out. Table 1 counts the events: 36 closures, 1 add-on (Steubenville, decided in 2003) and 1 conversion (Lorain, decided in 2011). The plant model takes the entry of the 24 new EAF mills in the sample as given; Section 4.3 models it. The decomposition of the emissions-intensity decline in Section 3.1 uses the whole panel, the excluded shops included.

3 Descriptive Evidence

3.1 Sources of the emissions-intensity decline

The EAF share of U.S. crude steel rose from 34% in 1985 to 71% in 2020, while the rest of the world stayed at 24% (Appendix Figure C.2). To see which plant decisions lowered emissions, I split the change in the capacity-weighted CO_2 intensity of the plants in the panel between 1985 and 2020 into new plants, closed plants, change at surviving plants

Table 1: Estimation sample and events, decisions 1985–2014

			Events				
Configuration	Plant-years	Sites	Closures	Add-ons	Conversions	Retirements	Idle years
Integrated works	518	21	6	1	1	0	11
Legacy hybrids	81	5	2	0	0	0	0
Hybrids after an add-on	4	1	0	0	0	1	0
EAF mills	2,160	100	28	–	–	–	19
All	2,763	122	36	1	1	1	30

Notes: Carbon-steel sites in the estimation scope. Plant-years: decisions in 1985–2014, each with its outcome in the next year; the decisions of 2015–2019 are held out. Legacy hybrids ran an EAF beside their blast furnaces from before 1985. Closure: the site is absent in the next year. Add-on: an EAF built beside the blast furnaces; conversion: the blast furnaces shut and the works becomes an EAF mill; retirement: a works with an add-on closes its blast furnaces. Idle years: capacity on the books, no output; counted as decisions to keep. A site counts once per configuration it passes through; the last row counts distinct sites. –: not open to the configuration. EAF: electric arc furnace.

and shifts among them, following Olley and Pakes (1996) extended to entry and exit (Appendix C.1). A plant's intensity mixes its routes: 2.0 tonnes of CO_2 per tonne for integrated capacity, and for EAF capacity its direct emissions plus its electricity times the CO_2 factor of the state's power sector in that year (EIA 2026c; EIA 2025c). The split therefore sees which plants exist, how large they are and how clean their electricity is, but not plants becoming more efficient in other ways, the technique channel of Shapiro and Walker (2018).

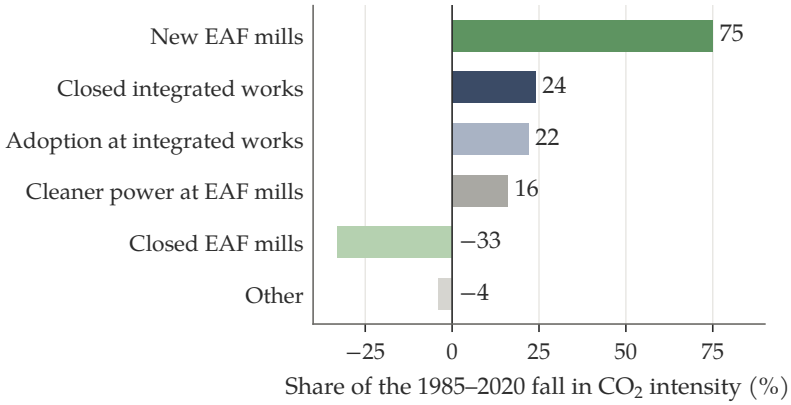
Intensity fell from 1.28 to 0.89 tonnes of CO_2 per tonne of capacity, by 31% (Figure 2). New plants, all of them EAF mills, account for 75% of the fall. Closures of the old works account for 24%, about a third as much. The three works that adopted the EAF account for 22%, and surviving EAF mills on a cleaner grid for 16%. This is the replacement of integrated works by EAF mills that Barnett and Crandall (1986) described and Collard-Wexler and De Loecker (2015) measured as a reallocation of output; here it is split by its effect on CO_2 intensity. Other forces that could have moved the industry, among them demand, imports, energy prices and environmental regulation, are discussed in Appendix C.6.

Closures of EAF mills have a negative share, −33%, because the mills that closed were cleaner than the 1985 average, not because closing them raised emissions. Together, the closures of both routes raised average intensity (−9%). With a fixed EAF intensity, and in every other variant, entry remains the largest part.

3.2 The adoption record

Only three old works adopted the EAF between 1985 and 2020, and none in one step (Figure 3; dates in Appendix Table A.2). Steubenville, owned by Wheeling-Pittsburgh, built its EAF beside its two running blast furnaces and retired one of them when the EAF started, keeping the other until 2008. After the owner's Chapter 11 filing, a federal loan guarantee of USD 250 million, approved in 2003, supported financing that included a USD 110 million EAF, which tapped its first heat, its first batch of steel, in 2004.

Figure 2: Decomposition of the fall in CO_2 intensity, 1985–2020



Notes: Shares of the 1985–2020 fall in the capacity-weighted CO_2 intensity of all plants in the panel, specialty and stainless shops included. Integrated capacity emits 2.0 tonnes of CO_2 per tonne; EAF capacity its direct emissions plus its electricity times its state's power-sector CO_2 factor. Cleaner power: the fall in the grid factors of surviving EAF capacity. Other: EAF steelmaking shops closed at surviving works plus shifts in capacity among surviving plants (−4% unrounded), set so that the integers add to 100. Variants in Appendix Table C.1. EAF: electric arc furnace.

Steelmaking continued during construction. The firm's annual report called the result "a hybrid steel producer".[2] The blast furnaces closed only in 2008. The EAF itself stood idle from 2010 to 2017 and ran again from 2018 (Figure 3, shaded years).

The two conversions came after years without steelmaking. Lorain's owner, Republic Technologies, filed for Chapter 11 in 2001, and the PBGC took over its pension plans in 2003. In 2005, Industrias CH bought the business and kept the blast furnaces running. Steelmaking stopped in 2008, and the works stood idle without closing. The owner decided to convert in 2011; the EAF started in 2013, with 0.46 of the old capacity, making bars, and stood idle from 2016 on. At Fairfield, U.S. Steel, solvent throughout, shut the blast furnaces in 2015, the year it announced the EAF, which started in 2020, after the estimation window, with 0.67 of the old capacity. I use these operating histories to set adoption capacities and construction times in the model (Section 4.1). The model allows an idle works to convert, and it takes Lorain's ownership history as given (Section 4.1).

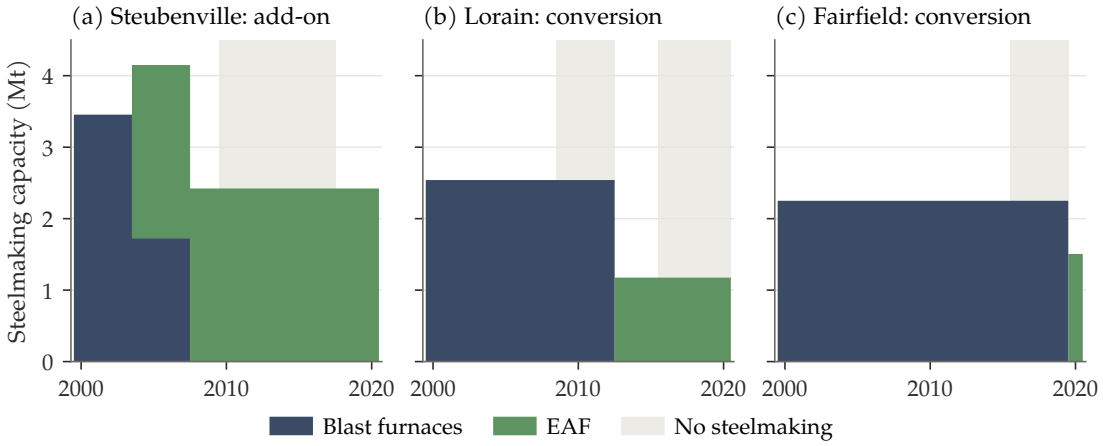
What the integrated firms said fits this record. In 306 annual reports of 23 integrated firms, fiscal years 1993–2024, no statement weighs adopting the EAF at an integrated works and rejects it, and no such plan predates Wheeling-Pittsburgh's. Integrated firms also built or bought little EAF capacity elsewhere: 14 cases over 1985–2021, most of them purchases (Appendix C.2). The model does not let the incumbents, the owners of the old works, own new mills.

3.3 Who bore shutdown costs

The integrated firms carried heavy pension and retiree health obligations, while the firms that ran EAF mills, with younger workforces, carried few (Cooney 2002). Among the decisions of owners with large gross obligations, unfunded pension and retiree

[2]Wheeling-Pittsburgh, 10-K for fiscal years 2003 and 2004 (SEC accessions 0000950128-04-000290 and 0000950152-05-002048).

Figure 3: The three adoptions: steelmaking capacity by route



Notes: Steelmaking capacity, Mt a year, of the three integrated works that adopted the EAF, by route. (a) Add-on: an EAF built beside running blast furnaces. (b), (c) Conversions: the blast furnaces shut and an EAF built in their place. Shaded: years without steelmaking while capacity stayed on the books. EAF: electric arc furnace.

health obligations averaged 199 USD per tonne of capacity over 1985–2014, against 260 USD per tonne to build an EAF steelmaking shop.

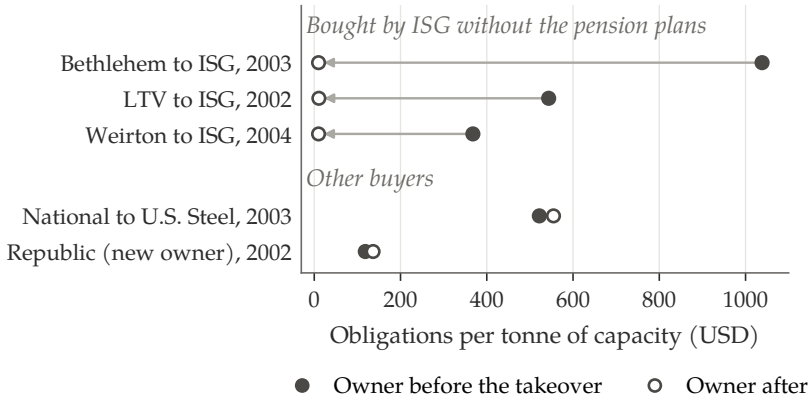
The obligations could change hands while the same blast furnaces kept running. Bankruptcy was the usual setting: of the 8 closures of integrated works in the estimation sample, 7 came at or near a bankruptcy of the owner. In the bankruptcies of 2002–2003, the PBGC took over the pension plans of LTV, Bethlehem, National and Weirton (Pension Benefit Guaranty Corporation 2026); the LTV plans alone cost it about USD 1.9 billion (Pension Benefit Guaranty Corporation 2002). Before the takeovers, the obligations of LTV and Bethlehem, 544 and 1,038 USD per tonne of capacity, exceeded the cost of a new EAF steelmaking shop (Figure 4). The International Steel Group bought their works without the pension plans, and its own obligations were 11 and 10 USD per tonne; at most of these works the same blast furnaces ran on. A buyer did not always shed the obligations: U.S. Steel, buying National's works, brought its own plans, and obligations per tonne did not fall (522 before, 555 after).

Cleaning up a site is the shutdown cost that uses real resources. The environmental accruals in the annual reports of 12 owners of integrated works average 9 USD per tonne of integrated capacity over works-years (Appendix Table A.1 lists them by owner); firms record only amounts they judge probable, so these are lower bounds. The cleanup falls due when the blast furnaces close. I use these accruals to construct the model's site-cleanup charge (Section 4.2).

3.4 Obligations and closure

The two observed adoptions give too little variation to study adoption directly, so the test is on closure. When a blast furnace needs relining, its owner chooses between relining it and closing it. I take the furnace-years in which a furnace at an integrated works of the estimation sample is due for relining, 1986–2015, and ask whether owners

Figure 4: Owners' obligations at the PBGC takeovers



Notes: Unfunded pension and retiree health obligations of the owner per tonne of its steelmaking capacity, USD, before (filled) and after (open) the year the works changed hands at a takeover of the owner's pension plans by the Pension Benefit Guaranty Corporation (PBGC). ISG: International Steel Group. The PBGC does not take over retiree health obligations.

with larger obligations closed less often. Obligations here are gross, before plan assets, unlike the model's unfunded measure.

Larger obligations go with fewer closures. Over the due furnace-years that closed or continued (Appendix Table C.2; design in Appendix C.4), the odds of closing a due furnace are lower by a factor of 0.73 per 100 USD per tonne of obligations (95% interval 0.59–0.88). The estimated ratio stays below one with the owner's bankruptcy state (0.80), with balance-sheet and works controls (0.82) and within owners over time (0.70). Being in bankruptcy predicts closure more strongly: Chapter 11 multiplies the odds of closing by 4.22 (2.00–8.82). The one comparison that could separate the two, a takeover of the pension plans without a bankruptcy in force, has 120 furnace-years and 2 closures, too few to detect an effect.

The association is not shown to be causal. Obligations change within an owner mainly through bankruptcy filings, sales and takeovers of pension plans, which are themselves signs of distress. This association is consistent with the shutdown-cost mechanism in Section 4.2, but does not identify its causal effect.

4 Model

The model is a dynamic model of an industry (Aguirregabiria, Collard-Wexler, and Ryan 2021): plants choose whether to adopt the EAF, close or keep running, and new mills choose whether to enter. A new mill is valued as an operating EAF mill is, and the output of plants and new mills moves the prices both face. Margins, building costs, shutdown costs, hazards and the discount factor are set outside the likelihood. The shock scale, the closure values, the fixed adoption cost and the entry costs are estimated from the owners' and entrants' choices.

4.1 The plant problem

Plants and configurations. A plant is a site i with steelmaking capacity K_i in tonnes a year (the subscript is dropped where clear). Throughout, t indexes years, a choices, y the segment (flat or long) and $r \in \{B, E\}$ the route (BOF or EAF). The site's *configuration* ζ records how its capacity is split between the routes: an integrated works ($\zeta = B$) has only blast furnaces, an EAF mill ($\zeta = E$) only EAFs, and the two hybrid configurations, a works with an add-on and the legacy hybrids that ran an EAF beside their blast furnaces in 1985, have both.

Choices. The owner of an integrated works can keep it running, add an EAF beside the running blast furnaces, convert it (shut the blast furnaces and build an EAF in their place) or close it. A hybrid can keep both routes, retire its blast furnaces or close. An EAF mill keeps running or closes. An idle site holds, restarts, converts if integrated, or closes. Closing is permanent. Table 2 lists the choices and their one-time payoffs.

Chance events. Three events arrive by chance, with hazards fitted outside the likelihood (Appendix D.2). With ν_{it} the campaign age of a works' main blast furnace (years since its last reline), a reline need arrives with probability $h^R(\nu)$ in each operating year. Keeping is then no longer open: the owner relines (paying C^R per tonne of BOF capacity), adds an EAF, converts or closes. An operating site is idled with probability h^I a year, paying C^I per tonne on idling and C^H per tonne in each idle year, with its campaign age standing still. A restart costs C^S per tonne. An operating EAF configuration replaces a furnace with probability h^F_t a year and pays C^F per tonne of its EAF capacity.

Timing, capacities and output. At the start of year t the owner observes the state and earns the year's flow payoff. An operating site is then idled, or a reline need arrives, with the probabilities above. The owner observes a shock to each choice open to it and chooses. Capacity is taken from the data, not chosen. An add-on builds EAF capacity of $\kappa_{\text{add}} = 0.70$ of the works' capacity and keeps 0.50 of it as BOF capacity. Output stays within the works' capacity, so the EAF produces on 0.50 of it and the blast furnaces on the rest. A conversion leaves an EAF mill of the share $\kappa_{\text{conv}} = 0.46$. Retiring a hybrid's blast furnaces leaves its EAF share. A conversion takes $T^{\text{conv}} = 2$ years to the first heat, with no steel meanwhile. An add-on takes $T^{\text{add}} = 1$ year while the blast furnaces run. These sizes and times are set from the observed adoptions (Section 3.2). A configuration's EAF and BOF capacity are denoted K^{EAF} and K^{BOF}, and its BOF capacity at the works' effective rate, the smaller of BOF capacity and blast-furnace supply, K^{eff}.

The state and expectations. The state of plant i in year t is $\omega_{it} = (n_{it}, z_t, t; x_i)$. The *node* n_{it} is the configuration ζ_{it} with whether the site operates, is idle or is under construction (and in which year), and the campaign age ν_{it} of a works with a blast furnace. Owners do not know future steel margins: they expect today's margins to drift back toward their long-run means at the rates seen in the data. Formally, the price state z_t collects

the deviations of the BOF flat, EAF flat and EAF long national margin indices from their means and follows a stationary vector autoregression, discretized into a Markov chain (Appendix B.4). Calendar time t indexes the dated cost path, held fixed after 2015. Owners know it and forecast the margins with the chain. The characteristics x_i are the plant's segment, its persistent cost, utilization and product-mix terms, and the obligation level $L_f \in \{0, 0.5, 1\}$, which classes the owner f as having no, moderate or large unfunded obligations per tonne of its capacity (Appendix D.1). The owner treats them as permanent. A run of years with unchanged characteristics is one spell.

Yearly profit. The flow payoff $\pi_t(n, z)$ of an operating site, in USD, is output q^r_{it} in tonnes times the net margin m^r_{it} in USD per tonne of each route present, less three costs of particular configurations:

$$\pi_t(n,z) = q^B_{it}\, m^B_{it}(z) + q^E_{it}\, m^E_{it}(\zeta, z) - \alpha\, o^B_t\, q^{\text{idle}}_{it} - C^A \max\{\nu - \bar{\nu}, 0\} K^{\text{BOF}} - h^F_t C^F K^{\text{EAF}}. \quad (1)$$

The last three terms are the iron-balance cost of an add-on hybrid (a share $\alpha = 0.5$ of the BOF non-material cost o^B_t on q^{idle}_{it}, the output of its idled blast-furnace capacity), the rising cost of an old campaign (C^A per tonne of BOF capacity per year beyond a campaign length $\bar{\nu}$) and the expected furnace-replacement outlay of an EAF configuration without blast furnaces. Output is not an explicit choice in the model: in words, each route produces its capacity, run at that year's national utilization rate of the route and adjusted by the plant's own persistent utilization factor, $q^r_{it} = U^r_t U^{\text{p}}_i K^r$, with K^r the capacity of route r (for the BOF at the effective rate), U^r_t the route's national utilization rate and U^{p}_i the plant's persistent utilization factor.

Margins, product restrictions and the technology path. The net margin of route r at plant i is

$$m^r_{it}(z) = \mu^r_{yt} + z^r + C^{\text{mat},r}_i - o^r_t - C^{\text{down},r}_{yt} + p^{\text{mix},r}_{it}, \quad (2)$$

where $\mu^r_{yt} + z^r$ is the national margin index of route r in segment y (its mean on the dated cost path plus the route's component of z), $C^{\text{mat},r}_i$ the plant's material-cost term, o^r_t non-material cost, $C^{\text{down},r}_{yt}$ the downstream cost of casting and rolling the steel into the base product and $p^{\text{mix},r}_{it}$ the price of the plant's own product mix over the base product (Appendix B.1 gives the prices and mixes, Appendix B.2 the costs). After an adoption the new EAF cannot sell the works' whole mix: in the configurations that adoption creates, and in the EAF mill left when a hybrid retires its blast furnaces, its output sells no tin plate and at most a share $\bar{w}_t$ of it is cold-rolled or coated, the share that EAF sheet mills reached by period (Appendix B). The excess above the cap sells as hot-rolled sheet. The EAF technology path is exogenous to plants and policy: a quality penalty on the EAF flat margin before 1989, when thin-slab casting brought EAF mills into sheet, the period-specific cap $\bar{w}_t$ and the trend in the entry cost (Section 4.3).

4.2 Payoffs and values

Building cost. An add-on or a conversion pays I^C per tonne of new EAF capacity, 260 USD per tonne, the cost of replacing an EAF steelmaking shop, in installments over construction. The works' casters, rolling mills and site are reused.

Shutdown costs. As in Ryan (2012), shutting down capacity can carry a one-time cost. Here, shutting a works' blast furnaces brings two bills. The first is the owner's obligations to its workers and retirees, D^C on conversion and on retiring a hybrid's blast furnaces and D^X on closure:

$$D^C_{it}(\zeta) = \bar{\lambda}\, L_f\, K^{\text{BOF}}(\zeta), \qquad D^X_{it}(\zeta) = d_X\, D^C_{it}(\zeta), \tag{3}$$

where $K^{\text{BOF}}(\zeta)$ is the BOF capacity of the configuration and $\bar{\lambda} = 206$ USD per tonne is the mean unfunded obligation per tonne of owners at $L_f = 1$. Closure pays D^X with $d_X = 1$, because the same obligations fall due whichever way the blast furnaces shut. An add-on pays nothing at the time, but its hybrid owes D^C on its remaining BOF capacity if it later retires the blast furnaces or closes. The second bill is site cleanup, $\mathcal{R}_{it}(\zeta) = C^{\text{rem}}_{it} K^{\text{BOF}}(\zeta)$, with the accrual per tonne C^{rem}_{it} from annual reports (Section 3.3). The obligations are transfers for the economy but costs to the owner. The cleanup uses real resources. Only closing, converting and retiring the blast furnaces pay them.

Obligations on conversion, and bankruptcy. First, the model assumes that converting brings the whole of D^C due. Shutdown triggers the obligations whatever form it takes, and converting ends the blast furnaces and most of their jobs, but no filing says what a firm would have owed had it converted while its plan stayed in force. If only part of the bill fell due, converting would cost the owner less at the time. The effect on the adoption gaps of Section 4.4 also runs through the continuation values. Second, the model has no bankruptcy or reorganization decision: observed changes of owner and of obligations enter through the data, and an owner in Chapter 11 carries its full obligation. The model takes ownership and the owners' obligation levels as given. It does not explain how bankruptcy, sales or financing change the opportunities to adopt.

Closure values and the fixed adoption cost. Closing pays S^X_{EAF} per tonne of EAF capacity and S^X_{BOF} per tonne of effective BOF capacity, before shutdown costs. A closure value is not a sale price but what closing is worth relative to running on, including scrap, land and whatever the margins leave out, such as fixed operating costs. Every add-on and conversion also pays a fixed adoption cost ϕ_C, in USD per adoption: what adopting the EAF costs or saves a works beyond the building cost, the forgone profit and the shutdown costs, not restricted in sign. One ϕ_C applies to add-ons and conversions and to all works, because there are only two adoptions.

Table 2: Choices, one-time payoffs and the configuration that follows

Configuration	Choice	One-time payoff $u_a(\zeta)$	Next year
Integrated works	Keep	0	Integrated works
	Reline (at an arrival)	$-C^R K^{\text{BOF}}$	Integrated works, $\nu = 0$
	Add-on	$-I^C \kappa_{\text{add}} K - \phi_C$	Add-on hybrid
	Convert	$-I^C \kappa_{\text{conv}} K - D^C - \mathcal{R} - \phi_C$	EAF mill ($\kappa_{\text{conv}} K$) after T^{conv} years
	Close	$S^X_{\text{BOF}} K^{\text{eff}} - D^X - \mathcal{R}$	Closed
Hybrid	Keep	0	Hybrid
	Retire blast furnaces	$-D^C - \mathcal{R}$	EAF mill
	Close	$S^X_{\text{EAF}} K^{\text{EAF}} + S^X_{\text{BOF}} K^{\text{eff}} - D^X - \mathcal{R}$	Closed
EAF mill	Keep	0	EAF mill
	Close	$S^X_{\text{EAF}} K^{\text{EAF}}$	Closed
Idle site	Hold	0	Idle site
	Restart	$-C^S K$	The operating configuration
	Convert (integrated)	As convert above	EAF mill after T^{conv} years
	Close	As close above	Closed

Notes: One-time payoff $u_a(\zeta)$ of choice a in configuration ζ, USD, and the next year's configuration; flow payoffs are added every year. Hybrid: an integrated works with an EAF beside its blast furnaces. A reline (rebuilding a blast furnace's lining) is open only when a reline need arrives. K, K^{EAF}, K^{BOF}: site, EAF and BOF capacity; K^{eff}: effective BOF capacity, the smaller of BOF capacity and blast-furnace supply. C^R, C^S: reline and restart cost per tonne; ν: years since the last reline. I^C: building cost per tonne of new EAF capacity; κ_{add}, κ_{conv}: EAF capacity of an add-on and a conversion, as shares of K; T^{conv}: years to a conversion's first heat; ϕ_C: fixed adoption cost. D^C, D^X: obligations due on shutting the blast furnaces and on closure; $\mathcal{R}$: site cleanup; S^X_{EAF}, S^X_{BOF}: closure values per tonne. EAF: electric arc furnace; BOF: basic oxygen furnace.

Shocks. Each choice a also receives a shock $\sigma_i \varepsilon_{ait}$, with ε_{ait} independent type-I extreme value and σ_i, in USD, the shock scale of the spell,

$$\sigma_i = \sigma K^{\text{med}} \Big(\frac{\bar{K}_i}{K^{\text{med}}} \Big)^{\rho}. \tag{4}$$

Here $\bar{K}_i$ is the capacity of the configuration in which the spell starts, its BOF part at the effective rate, and K^{med} is the median of $\bar{K}_i$ over the spells of the sample, 1.06 million tonnes. The shock scale σ is in USD per tonne at the median capacity, and ρ is the elasticity of σ_i with respect to capacity: the shock per tonne is the same at every size when $\rho = 1$ and falls with capacity when $\rho < 1$. The shocks capture reasons for a choice that the data do not record, such as an offer from a buyer for the site or a manager's view of a project. They are why the same plant in the same state does not always make the same choice.

Choice values. For the menu $\mathcal{A}$ of choices open at node n, the logit expected maximum, with γ Euler's constant, is

$$\bar{v}_t(\mathcal{A} \mid n, z) = \sigma_i \Big(\gamma + \log \sum_{a \in \mathcal{A}} \exp\big(v_t(a \mid n, z) / \sigma_i \big) \Big), \tag{5}$$

with choice values

$$\begin{aligned} v_t(a \mid n,z) &= u_a(n) + \beta\,\mathbb{E}\big[V_{t+1}(n'(a),z') \mid z\big] \quad \text{for } a \neq \text{close},\\ v_t(\text{close} \mid n,z) &= u_{\text{close}}(n), \end{aligned} \tag{6}$$

where $u_a(n)$ is the one-time payoff of Table 2, $n'(a)$ the node that follows a, z' next year's price state and β the annual discount factor, fixed at 0.85 (Section 5.1).

The value of a site and the choice probabilities. Let $\mathcal{A}^{\text{reg}}(n)$ be the regular choices of an operating node and $\mathcal{A}^{\text{arr}}(n)$ the choices at a reline arrival. The value of an operating site at the start of year t is

$$\begin{aligned} V_t(n,z) = \pi_t(n,z) &+ h^I\big(-C^I K + \beta\,\mathbb{E}[V_{t+1}(n^{\text{idle}},z') \mid z]\big)\\ &+ (1-h^I)\Big(h^R(v)\,\bar{v}_t\big(\mathcal{A}^{\text{arr}}(n) \mid n,z\big) + \big(1-h^R(v)\big)\,\bar{v}_t\big(\mathcal{A}^{\text{reg}}(n) \mid n,z\big)\Big), \end{aligned} \tag{7}$$

where n^{idle} is the idle node of the same configuration and $h^R = 0$ without a campaign clock. Within a menu $\mathcal{A}$ the probability $\Pr_t(a \mid \mathcal{A},n,z)$ of choosing a is the logit of the scaled choice values $v_t(a \mid n,z)/\sigma_i$, Equation (20). Appendix D.3 gives the values of idle and construction nodes and the solution.

4.3 Entry and prices

In each decision year $t = 1985,\ldots,2013$ and segment y, a pool of $\bar{N} = 10$ potential entrants (a normalization: only the entry cost net of $\sigma_{\text{ent},y}\log\bar{N}$ is identified, see below) each considers one EAF mill, which produces after $T^{\text{ent}} = 2$ years. A project knows its capacity K, drawn from the observed entrant sizes of its segment, when it decides and learns its utilization factor U^{p}, drawn from the observed entrants' utilization factors, only after it builds (Appendix D.3). The value of entry per tonne, before the entry cost, is

$$V_y^{\text{ent}}(t,K) = -I^C + \frac{\beta^{T^{\text{ent}}}}{K}\,\mathbb{E}_t\mathbb{E}_{U^{\text{p}}}\big[V_{t+T^{\text{ent}}}(n^E,z')\big], \tag{8}$$

where V is the value (7) of an operating EAF mill n^E of the segment, with no obligations, the entrants' product mix and the furnace-replacement outlay. The model treats new mills as independent of the incumbents (Section 3.2).

A potential entrant builds if this value exceeds its entry cost $F_y^{\text{ent}} + \tau(t-2000)$ plus a logistic shock of scale $\sigma_{\text{ent},y}$ per tonne. The entry cost covers what a new mill needs beyond the steelmaking shop and other barriers to entry. The trend τ absorbs the diffusion of thin-slab casting and the falling number of good sites. With Λ the logistic distribution function and $\mathbb{E}_K$ the mean over the five equal-weight quantiles of the size distribution of K, the probability $\Pr_y^{\text{ent}}(t)$ that a project enters is

$$\Pr_y^{\text{ent}}(t) = \mathbb{E}_K\,\Lambda\Big(\frac{V_y^{\text{ent}}(t,K) - F_y^{\text{ent}} - \tau(t-2000)}{\sigma_{\text{ent},y}}\Big). \tag{9}$$

Only $F_y^{\text{ent}} - \sigma_{\text{ent},y} \log \bar{N}$ is identified, so the entry cost is stated for the pool of 10. The entrants' shock is tied to the plant model's at the entrants' capacity (Section 5.1). The logit entry probability follows Collard-Wexler (2013) and the construction lag Kalouptsidi (2014). Entrants are anonymous.

When a policy changes domestic output by ΔX_{yt}, the price of segment y moves along the net demand facing U.S. producers, with slope the denominator of

$$\Delta p_{yt} = -\frac{\Delta X_{yt}}{\left(\eta_y^d Q_{yt} + \eta_y^m M_{yt}\right)/p_{yt}}, \tag{10}$$

where p_{yt} is the segment's price, Q_{yt} apparent consumption, M_{yt} imports, η_y^d the demand elasticity (0.20 for flat and 0.25 for long products, chosen in Appendix C.5) and η_y^m the import-supply elasticity (1.48 and 1.30, estimated). Input prices, including scrap, do not respond in the main experiments.

In the policy experiments, prices respond to the plants' and entrants' choices: a grant changes how much steel is made, and the resulting prices change who adopts, closes and enters. Plants and entrants foresee this price path, which shifts every margin one for one, but not the random swings in margins around it, and the experiment finds the price path that is consistent with the output it produces. In estimation, by contrast, owners take observed prices as given and expect them to follow the estimated chain, and the trend τ in entry costs stands in for the way entry lowers prices. Because τ is estimated on observed prices, which already reflect entry, letting prices respond in the experiments may count part of this feedback twice and overstate how much prices damp entry.

4.4 The adoption gap

For an integrated works i in year t, at an operating node n with configuration B and price state z, and for $a \in \{\text{add-on}, \text{convert}\}$, the *adoption gap* is

$$\Gamma_{it}(a) \equiv \max\{v_t(\text{keep} \mid n, z),\, v_t(\text{close} \mid n, z)\} - v_t(a \mid n, z), \tag{11}$$

the one-time payment, in USD, that would make choice a worth as much to the owner as the better of keeping and closing the works, before shocks. I report it per tonne of site capacity. The gap compares the whole future of each choice, not one year's profits: keeping includes the blast furnaces' remaining profits and the option to close later; converting includes the years without steel, the smaller shop and the cap on its high-end products.

I group the parts of the gap into three categories. Section 5.3 reports their shares of the gap net of two pieces unrelated to adoption. The *price of the technology* is what the new furnace costs and can make: the building cost, the size of the shop and the cap on its high-end products, which a new mill faces too. The *forgone profit* is what the running blast furnaces still earn, which only an old works gives up by adopting. The *structural frictions* are specific to adopting at an old works: the shutdown obligations and the fixed adoption cost. The first two are real costs, which a grant can pay for but not remove.

The grant and carbon-price experiments hold the frictions fixed. The pension-takeover experiment changes who bears the obligations.

5 Estimation and Results

The plant model has five parameters to estimate: the fixed adoption cost ϕ_C, the shock scale σ and its elasticity ρ, and the closure values S^X_{EAF} and S^X_{BOF}. Entry adds the entry costs of flat and long mills and their trend. All other inputs, among them the margins, the building cost, the shutdown obligations and the discount factor, are set outside the likelihood (Appendix Table D.1), and the intervals below take them as given. Values are in USD of 2005, and intervals are 95% profile-likelihood intervals unless stated.

5.1 Identification

I estimate the plant model by full-solution maximum likelihood with a nested fixed point (Rust 1987). For each candidate $\theta = (\phi_C, \sigma, \rho, S^X_{\text{EAF}}, S^X_{\text{BOF}})$, the model is solved for every plant, and each decision contributes the probability of the observed choice, mixed over the menus because relines and idling are chance events (Appendix D.3). The likelihood counts three fewer decisions and one more closure of an EAF mill than Table 1, because two sites' timing is recoded (Appendix A.2). The plant log likelihood sums over the 2,760 decisions at 122 sites,

$$\ell^P(\theta) = \sum_i \sum_t \log \Pr\big(a_{it} \mid n_{it}, z_t, t; \theta\big). \tag{12}$$

A new mill is the plant model's EAF mill, so entry and the plants' choices share the shock scale, the discount factor, the price process and the value of a mill. With N_{yt} the observed entries in segment y and year t, the entry log likelihood is the binomial

$$\ell^{\text{ent}}(\theta, F^{\text{ent}}, \tau) = \sum_y \sum_t \log\Big[\binom{\bar{N}}{N_{yt}} \Pr^{\text{ent}}_y(t)^{N_{yt}} \big(1 - \Pr^{\text{ent}}_y(t)\big)^{\bar{N} - N_{yt}}\Big], \tag{13}$$

over the 24 entries of new mills into the sample's scope (13 flat, 11 long), and I estimate entry and the plant model together,

$$\ell(\theta, F^{\text{ent}}, \tau) = \ell^P(\theta) + \ell^{\text{ent}}(\theta, F^{\text{ent}}, \tau), \qquad \sigma_{\text{ent},y} = \sigma\Big(\frac{K^{\text{ent}}_y}{K^{\text{med}}}\Big)^{\rho - 1}, \tag{14}$$

The entrants' decisions after entry are already plant decisions in ℓ^P, so nothing is counted twice. The entrant's shock per tonne is the plant model's at the entrant's capacity K^{ent}_y, the mean capacity of the observed entrants of segment y, and ρ is held at the plant model's estimate. The plant parameters, the adoption gap and the policy experiments rest on the plant model, with entry costs estimated given it. The intervals of the entry costs come from the joint estimate (Table 3).

Different patterns in the data inform the parameters. The 29 closures of EAF mills, set against their margins, which vary across years and segments, inform σ and S^X_{EAF}:

how strongly closures respond to margins shows how much choices vary for reasons the model does not see, and how often mills close at given margins shows what closing is worth. The 8 closures of integrated works and hybrids inform S^X_{BOF} in the same way, given the shutdown obligations. Closures by size inform ρ: most integrated works are above the median capacity and most EAF mills below it. The fixed adoption cost rests on the choices to adopt: what adopting would have earned varied across works and years with steel margins, the works' size and the owner's obligations, and the many years in which works could have adopted and did not bound ϕ_C from below, while the 2 adoptions bound it only weakly from above. One ϕ_C covers both forms of adoption: the data cannot separate a cost of the add-on from one of converting, although the model computes different total gaps for the two.

Choice data alone do not identify the discount factor without further restrictions on payoffs (Rust 1994; Magnac and Thesmar 2002). A higher β raises the value of keeping a works running, and the model answers with larger closure values and a larger ϕ_C. I chose β by a rule fixed before estimation, on a grid of values: a value is plausible if both closure values (for integrated works, net of the obligation scale) lie within a factor of 3 of the median price buyers paid for going plants of that kind (Appendix C.3). Only $\beta = 0.85$ passes. Below it the closure value of EAF mills is negative, above it that of integrated works net of the obligations exceeds three times the median price. Appendix D.4.1 gives the grid and the estimates at each value.

A recovery test on simulated panels, against criteria fixed beforehand, finds little bias and adequate coverage for the plant model and the joint estimate. With two adoptions, the interval for ϕ_C is wide and its upper end weakly informed (Appendix D.4.2).

5.2 Estimates

I estimate the fixed adoption cost at $\phi_C = 0.77$ billion USD per adoption (Table 3), about 414 USD per tonne of new EAF capacity against a building cost of 260. Engineering studies estimate what cleaner steelmaking costs to build and run, for routes from hydrogen-based direct reduction to carbon capture (Vogl, Åhman, and Nilsson 2018; Wu et al. 2025; Atouife, He, and Jenkins 2026; Jaramillo et al. 2023; Jordan et al. 2025). The fixed adoption cost is what owners' choices reveal on top of such costs at an old works. Its profile interval runs from 0.21 to 1.96 billion (113 to 1062 per tonne), and the joint estimate gives 0.72. The data reject a model without a fixed adoption cost (likelihood-ratio statistic 9.27, p-value 0.002), but the test rests on few events, since only two works adopted. The profile interval assumes that shocks are independent across plants. Because the plants of one firm may share shocks, I also resample whole firms with their plants, a pairs cluster bootstrap (Cameron and Miller 2015), and re-estimate the model in each draw (Appendix D.4.3). This gives a wider percentile interval, 0.42 to 12.3 billion, because 11 of its 100 samples contain no adoption and say little about the upper end of ϕ_C. Among the 89 samples with an adoption it runs from 0.40 to 1.83 billion.

At the median capacity the shock scale is $\sigma = 158$ USD per tonne of capacity, and its

Table 3: Estimates of the plant model and of entry

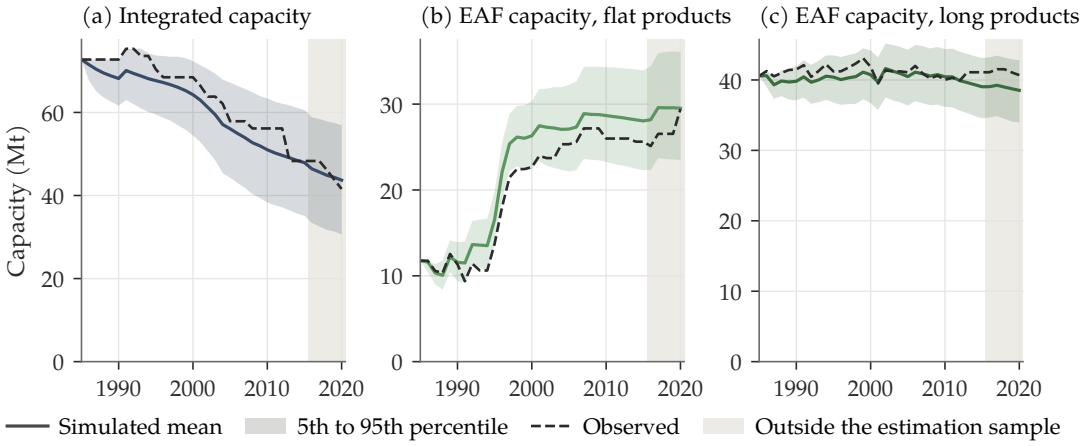
	Baseline estimate	Joint estimate with entry
A. Plant model		
ϕ_C, USD bn per adoption	0.77 [0.21, 1.96]	0.72 [0.19, 1.77]
ϕ_C, USD per tonne of new EAF capacity	414 [113, 1062]	391 [101, 959]
σ, USD per tonne at median capacity	158 (49)	154 [109, 229]
ρ, elasticity of σ_i to capacity	0.59 (0.20)	–
S^X_{EAF}, USD per tonne	96 (84)	100 (53)
S^X_{BOF}, USD per tonne	475 (47)	476 (46)
B. Entry		
$F^{\text{ent}}_{\text{flat}}$, USD per tonne	–	805 [570, 1204]
$F^{\text{ent}}_{\text{long}}$, USD per tonne	–	1059 [765, 1556]
τ, USD per tonne a year	–	13.7 [6.3, 23.5]
C. Tests and fit		
LR statistic, $\phi_C = 0$ (χ^2_1 p-value)	9.27 (0.002)	9.59 (0.002)
LR statistic, $\tau = 0$	–	12.3
LR statistic, σ_{ent} tied to σ (p-value)	–	0.88 (0.35)
LR statistic, $\rho = 1$ (p-value)	4.89 (0.027)	–
Log likelihood: plant decisions	−398.90	−398.90
Log likelihood: entry	–	−46.41
Plant decisions; entries	2,760	2,760; 24
D. Expected (observed) events, decisions 1985–2014		
Closures of EAF mills	27.0 (29)	26.9 (29)
Closures of integrated works and hybrids	7.1 (8)	7.1 (8)
Add-ons	1.7 (1)	1.7 (1)
Conversions	0.1 (1)	0.1 (1)
Retirements of a hybrid's blast furnaces	1.4 (1)	1.3 (1)
Idlings	9.0 (9)	9.0 (9)
Restarts from idle	11.3 (7)	11.4 (7)
Relines	38.9 (33)	38.9 (33)

Notes: Maximum-likelihood estimates, $\beta = 0.85$; brackets: 95% profile-likelihood intervals; parentheses: standard errors, or p-values where stated. ϕ_C: fixed adoption cost of an add-on or conversion, per adoption and per tonne of new EAF capacity (1.85 Mt per adoption on average). σ: choice-shock scale per tonne at the median capacity; ρ: elasticity of the total shock scale σ_i with respect to capacity. S^X_{EAF}, S^X_{BOF}: closure values per tonne of EAF and effective BOF capacity. $F^{\text{ent}}_{\text{flat}}$, $F^{\text{ent}}_{\text{long}}$: entry costs of new flat- and long-product mills; τ: their trend per year from 2000. Column 2: joint estimate with entry, the entrants' shock scale σ_{ent} tied to σ and ρ held fixed. LR: likelihood-ratio statistic. Panel D: expected counts (observed in parentheses); idlings match by construction. –: not in the model. EAF: electric arc furnace; BOF: basic oxygen furnace.

elasticity with respect to capacity is $\rho = 0.59$, so the shock per tonne falls with capacity. The data reject a shock proportional to it (likelihood-ratio statistic 4.89, p-value 0.027). The closure value of an EAF mill is $S^X_{\text{EAF}} = 96$ USD per tonne, below the median price of EAF mills in the deals, 201. A deal price is what a buyer pays to run a mill on, so it is an upper reference for what closing is worth. Since the deals also chose β, they are no independent check. The closure value of an integrated works is $S^X_{\text{BOF}} = 475$ USD per tonne, before shutdown costs. Besides scrap and land, its level can reflect fixed operating costs that closing avoids and the margins leave out (Appendix B.3).

Beyond the steelmaking shop, a new flat-product mill costs 805 USD per tonne to enter and a long-product mill 1059, rising by 13.7 USD per tonne a year. The trend is needed (likelihood-ratio statistic 12.3). Freeing the entrants' shock scale from the plant model's gives an entrants' shock scale 0.43 times the plant model's (likelihood-ratio statistic 0.88, p-value 0.35), so the entry data do not reject the tie, which anchors the

Figure 5: Installed capacity by route and product: observed and simulated



Notes: Installed steelmaking capacity, idle capacity included, Mt, at the carbon-steel sites of the estimation scope; EAF capacity at integrated works counts with the works' product. Simulated: mean and 5th to 95th percentile over 4,000 histories of the plant model at the baseline estimate without policy, from the 1985 state; new mills enter in the years they entered in the data. Shaded: 2016–2020, outside the estimation sample; these paths use inputs extended to 2020 with the parameters fixed and are not forecasts from the 2015 state. EAF: electric arc furnace.

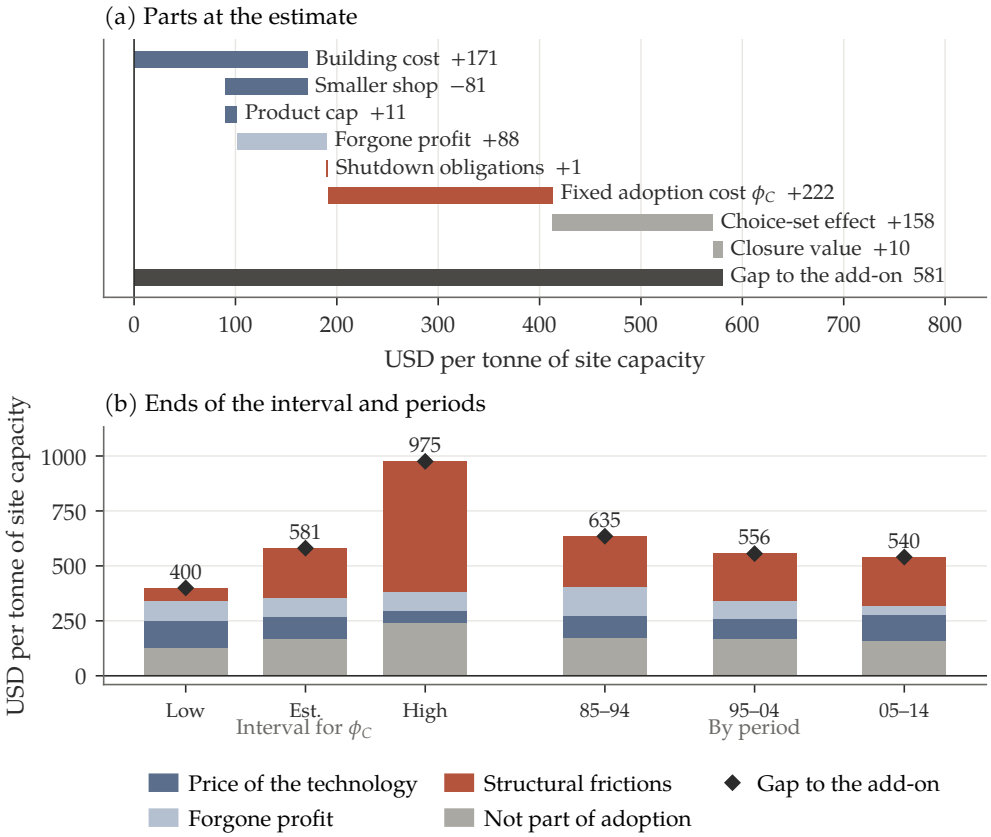
response of entry to a grant to the plants' choices rather than to 24 entries. The policy experiments use the entry costs estimated given the plant model, 830 and 1089 USD per tonne with a trend of 14.0 a year, inside the joint intervals of Table 3.

On the observed state paths the model expects 27.0 closures of EAF mills against 29 observed and 7.1 closures of integrated works and hybrids against 8 (Table 3, panel D). For adoptions, the choice probabilities sum to 1.7 add-ons against 1 observed and 0.1 conversions against 1. The total, 1.8 against 2, is close, but ϕ_C is estimated from these same decisions, and the split between the forms is poorly matched. In the held-out decisions of 2016–2019 the model expects 0.3 conversions against 1, and 2.3 closures of EAF mills against 2. These few events give limited evidence on the conversion response, on which the conversion grant of Section 6 particularly depends. The direction of any error in its cost is not established. Entry fits the totals but not their timing, since the linear trend cannot follow the wave of thin-slab mills of the 1990s (Appendix D.4.6). Figure 5 compares the capacity by route and product that the model simulates from 1985 with the observed capacity.

5.3 The adoption gap decomposed

The gap of Section 4.4 has seven parts that can be switched off: the building cost, the forgone profit, the shutdown obligations, the smaller shop, the product cap, the fixed adoption cost and the closure value. Because the parts interact, I use a Shapley decomposition: the part $c_{it}^{k}(a)$ is the effect on the gap of switching part k on, with the model solved again at the estimated parameters, averaged over all orders (Shorrocks 2013) (Appendix D.4.4). With $b_{it}(a)$ the gap with every part switched off, the parts add

Figure 6: The adoption gap and what it is made of



Notes: Shapley decomposition of the adoption gap of the add-on, Γ_{it}(add-on): the one-time payment that would make building an EAF beside the running blast furnaces as good to the owner as the better of keeping and closing. USD per tonne of site capacity; mean over the 507 works-years of running integrated works with an observed decision, 1985–2014. Smaller shop: the add-on shop is smaller than the works; product cap: the cap on the new EAF's high-end products; choice-set effect: a works draws a choice shock for each choice. (b) Low, Est., High: ends and estimate of the 95% interval for the fixed adoption cost ϕ_C (0.21, 0.77 and 1.96 billion USD); diamond: the gap. EAF: electric arc furnace.

up exactly:

$$\Gamma_{it}(a) = b_{it}(a) + \sum_{k=1}^{7} c_{it}^{k}(a). \tag{15}$$

Two pieces are not part of adoption and are netted out: $b_{it}(a)$, which for the add-on reflects only that a works draws a shock for each of its choices, and the closure value, which belongs to closing. I report the remaining parts as shares of this *net gap*, in the three categories of Section 4.4, as means over the 507 works-years in which an integrated works was running and made an observed decision in 1985–2014.

Of the mean gap to the add-on, 581 USD per tonne of site capacity (Figure 6), 168 is not part of adoption, which leaves a net gap of 413. At the estimate, the structural frictions are the largest category, 223 USD per tonne or 54% of the net gap, almost all of it the fixed adoption cost of 222. The fixed adoption cost runs from 58 to 591 USD per tonne across the interval for ϕ_C, so the frictions are 22% of the net gap at its lower end and 81% at its upper end. The price of the technology is 102 USD per tonne (25%): the building cost of 171, the largest single part after the fixed adoption cost, less 81 because

the add-on shop is smaller than the works, plus the product cap. The forgone profit is 88 (21%). The building cost is supplied from outside, but its part of the gap is computed in the model and depends on the estimated parameters too.

The add-on's gap is smaller than the conversion's in every works-year (means and shares by interval end and period in Appendix Table D.4). Under the model's assumption that converting brings the shutdown obligations due, they add 134 USD per tonne to the gap to convert, 20% of its net gap of 657, and little to the gap to the add-on, which leaves the blast furnaces running and pays nothing when it is built. The obligations therefore steer owners to the add-on rather than away from the EAF.

The forgone profit fell from 131 USD per tonne in 1985–94 to 81 in 1995–2004 and 41 in 2005–14, as the flat-product margins of integrated works fell toward those of EAF mills, and the gap fell from 635 to 540. Both observed adoptions came after the first period. The forgone profit also rests on the calibrated EAF margin, which reported profits check only roughly: a higher margin lowers it and, as Section 5.4 shows, raises ϕ_C.

The fixed adoption cost ϕ_C is an average across works and net of any unmodeled benefit of adopting, such as reusing the site's casters, rolling mills, customers and permits. Its components cannot be told apart. Candidates include costs the model prices at market or leaves out: the stranding of captive ore mines and coke batteries when the blast furnaces stop; the time customers take to re-qualify the new shop's steel; and permitting, if adding an EAF counts as a major modification that brings an old works under the stricter rules for new sources, which deter such changes at old plants (Stavins 2006). The estimate absorbs whatever the model leaves out, in either direction: a left-out benefit of the EAF, such as the flexibility to cut output in a downturn, lowers it below the true cost, and a left-out obstacle, such as financing a large project in distress, raises it.

5.4 Robustness

Margins. Appendix Table D.5 re-estimates the plant model, one setting at a time. The settings that move ϕ_C most concern the EAF margin, and each fits the choices worse than the baseline (-398.9). Pricing the EAF charge from scrap and pig iron from 1991 gives $\phi_C = 1.25$ billion, the engineering finishing cost for long products gives 1.35, and raising the EAF flat margin by 40 and 80 USD per tonne gives 1.30 and 2.02. A better EAF margin makes adopting more attractive, so a larger ϕ_C is needed to hold adoptions at two, and a larger σ to keep EAF closures on the data.

Obligations and capacity. Counting obligations before plan assets mainly moves the closure value: the obligation scale is 520 USD per tonne instead of 206, S^X_{BOF} rises from 475 to 664, absorbing most of the larger bill, and ϕ_C is 1.21 billion. Lifting the closure obligation of owners from two years before a Chapter 11 filing to their emergence gives $\phi_C = 0.99$ and a lower likelihood (-401.1), as the model then predicts too many closures among distressed owners. EAF capacity read from the directories' printed capacities

gives $\phi_C = 0.77$ billion, inside the baseline's interval.

Leaving out an adoption. Without Lorain's conversion ϕ_C rises to 0.98 billion (95% interval 0.27 to 2.52), and without Steubenville's add-on to 0.89 (upper end 2.33): with one adoption left, a larger cost explains why the other works did not adopt. The lower end stays above zero but rests on one or two events (Appendix D.4.6).

What survives. Re-running the gap decomposition and the grant comparison on these specifications leaves the ordering of the gaps unchanged. The obligations raise the gap to convert by 132 to 136 USD per tonne on the unfunded basis, but add little to the gap to the add-on. The building cost and the forgone profit are about half of the net gap or more only at the baseline EAF margin (48 to 59 percent). In the margin specifications the fixed adoption cost dominates. At the abatement target, the conversion grant is the most expensive of the three eligibility rules in every specification, and the adoption grant is the cheapest in seven of the nine, with the open grant slightly cheaper in the others. The model without idling, relines and time to build is in Appendix D.4.7.

6 Policy Counterfactuals

6.1 Instruments at equal abatement

I compare grants that pay for new EAF capacity under different eligibility rules, and use a carbon price as the benchmark. Beyond the CO_2 externality, which a carbon price corrects directly, a grant for adoption needs a market failure in investment itself to be the better instrument (Allcott and Greenstone 2012). Comparisons of instruments at a given emissions target find an emissions price cheaper than subsidies for clean electricity (Fischer and Newell 2008; Abrell, Rausch, and Streitberger 2019). The comparison here is among grants, with the carbon price as the reference.

Every policy starts in 1985, is expected to be permanent and is foreseen by plants and entrants. It is compared with the model's path without policy, the *fitted path*, which gives 1.56 adoptions in 1985–2014 against 2 observed. CO_2 avoided is counted over 1985–2015 as the fall in domestic emissions plus the emissions of the imports a policy displaces, credited at 1.2 tonnes per tonne. EAF steel is counted at 0.4 tonnes of CO_2 per tonne and integrated steel at 2.0. All policies are compared at one abatement target, 311 Mt of CO_2, the abatement that a carbon price of 21.3 USD per tonne would have bought on the fitted path. That price is the equivalent of the joint contribution of the EAF route's technology and input prices to the transition (Appendix E.2), and I use it to set one common target. Values are in USD of 2005, and results use the plant model's estimate with the entry costs estimated given it.

The *open grant* pays a level per tonne of new EAF capacity to every plant that builds it, new mills included. The *adoption grant* pays only old works that add an EAF or convert, and the *conversion grant* pays only old works that convert, building the EAF

Table 4: Cost-effectiveness at equal abatement

		Grants					
	Carbon price	Adoption	Conversion	Conditional	Open	Retention	Takeover
A. Abatement and response							
CO_2 avoided (Mt)	311	311	311	311	311	311	(188)
Level	21	351	1,098	416	137	64	(100)
Extra adoptions	+1.2	+8.9	+8.0	+4.7	+1.9	+8.8	(0.0)
Extra entries	+5.9	+0.6	+2.8	+3.3	+18.3	−0.5	(+4.6)
B. Public spending (USD per tonne of CO_2)							
Budget per tonne	–	27.5	49.5	33.4	28.1	108.9	(22.5)
Share ι (%)	–	13	1	21	54	32	–
Added investment	–	23.8	49.0	26.4	12.8	74.1	–
C. Welfare (USD per tonne of CO_2)							
Welfare cost	5.6	21.4	15.3	−7.4	−29.0	24.6	(−5.7)
Without shocks	49.0	89.1	87.2	77.2	98.8	86.5	(41.6)
Break-even value	6.2	22.6	19.8	−5.2	−29.3	26.0	(−5.9)
Without shocks	54.7	98.5	97.7	85.7	114.8	95.2	(46.6)

Notes: Each instrument at the level that avoids 311 Mt of CO_2 over 1985–2015, domestic emissions plus displaced imports at 1.2 tonnes per tonne. Grants: adoption, conversion, conditional, open and site-retention grants; takeover: obligation takeover. Level: carbon price per tonne of CO_2, grants per tonne of capacity (site retention per year), takeover in percent of the obligations; in parentheses, the takeover at its largest level, short of the target. Extra adoptions (1985–2014) and entries of new EAF mills: changes from the path without policy. Budget per tonne: payments over CO_2 avoided, undiscounted; share ι: the share of it the grant would pay on the path without policy; added investment: payments for added investment, budget times $(1-\iota)$. Welfare cost: the loss of consumers, producers and government per tonne, undiscounted; without shocks: producer surplus without the expected logit choice shocks. Break-even value: the social cost of carbon at which net welfare, discounted at 2%, is zero. EAF: electric arc furnace.

and retiring the blast furnaces in one decision.[3] Four other instruments complete the comparison: a *conditional grant*, paid when an old works retires its blast furnaces; a *site-retention grant*, paid each year an old site runs an EAF; an *obligation takeover*, in which a public fund pays a share of the shutdown obligations when they fall due; and the *carbon price*, on both routes. Each grant pays on every qualifying tonne, including those that would have been built without it.

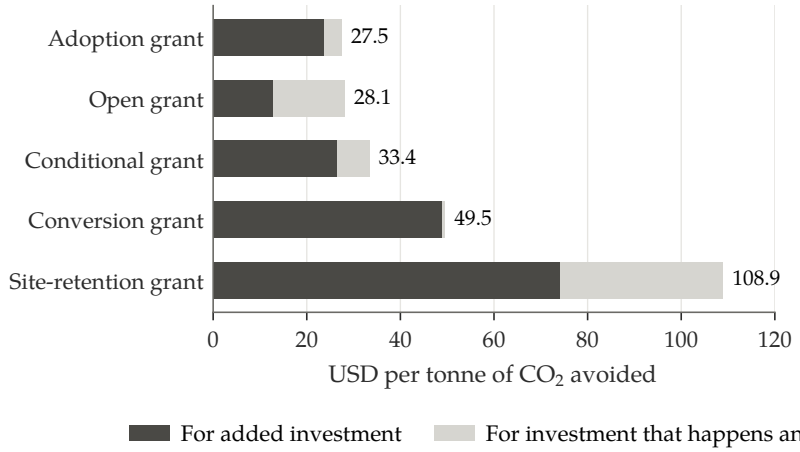
For a grant of level G per tonne, let K^0 be the qualifying capacity on the fitted path, ΔK the capacity the grant adds and ΔCO_2 the CO_2 it avoids. Public spending per tonne of CO_2 avoided is

$$C^G = \frac{G\left(K^0 + \Delta K\right)}{\Delta CO_2} = \frac{G\,\Delta K}{\Delta CO_2} \cdot \frac{1}{1-\iota}, \qquad \iota = \frac{K^0}{K^0 + \Delta K}. \tag{16}$$

Here ι is the share of the payments made for investment that would have happened anyway, and $G\,\Delta K/\Delta CO_2$ the payment per tonne of CO_2 for the investment the grant adds. Spending per tonne is low when the qualifying plants respond strongly to the grant, when each added tonne of capacity avoids much CO_2, and when little is paid for what happens anyway. For an agency with a fixed budget it is the relevant cost: the cheaper each tonne, the more abatement the budget buys.

[3] Butters, Dorsey, and Gowrisankaran (2025) evaluate up-front subsidies, as a share of capital cost, to battery storage in an equilibrium model of the California electricity market in which the option value of waiting for cheaper capital delays adoption.

Figure 7: Public spending per tonne of CO_2 avoided, by what it pays for



Notes: Each grant at the level that avoids 311 Mt of CO_2 over 1985–2015, as in Table 4; USD of 2005 per tonne of CO_2 avoided, undiscounted. Bar length: the budget per tonne. Investment that happens anyway: the investment the grant pays for on the model's path without policy.

At the abatement target, restricting eligibility cuts the share paid for investment that would have happened anyway from 54 percent under the open grant to 13 percent under the adoption grant and 1 percent under the conversion grant (Table 4 and Figure 7). Spending per tonne of CO_2 is nearly the same for the open and the adoption grant, 28.1 and 27.5 USD, and 49.5 for the conversion grant. The obligation takeover cannot reach the target, and the carbon price pays nothing and raises 64.0 billion USD. Section 6.2 explains these numbers, and Section 6.4 compares the instruments by welfare.

6.2 Payments, investment and emissions

Equation (16) splits the comparison of the adoption and open grants into three factors (Table 4). The first is the grant needed: old works respond weakly to grants, so the adoption grant must pay 351 USD per tonne of new EAF capacity to reach the target, against 137 for the open grant. The second is the total CO_2 avoided per tonne of additional qualifying EAF capacity: about 15 tonnes over 1985–2015 under the adoption grant and 11 under the open grant. It counts every change the policy brings about, in closures, in the output of other plants, in imports and in consumption, and depends on when the capacity is built and how long it runs. Together the two factors put spending on added investment at 23.8 USD per tonne of CO_2 under the adoption grant and 12.8 under the open grant. The third is the share paid for what happens anyway, 13 percent of the adoption grant's payments against 54 percent of the open grant's. The open grant's larger share slightly more than offsets its lower spending on added investment.

The adoption grant buys add-ons, the owners' preferred form because an add-on leaves the blast furnaces running and does not bring the shutdown obligations due: almost all of the adoptions it adds are add-ons. The open grant draws in 18.3 new mills, which lower the flat price (Appendix Figure E.1) and push out other domestic output: 0.87 of the entrants' extra flat output replaces other domestic output, and 6.7 more EAF

mills close. Most of the displaced output is from blast furnaces, but the part from other EAF mills saves almost no CO_2. The entry it pays for anyway is the entry that the EAF route's technology and input prices brought about.

Full conversions are rare on the fitted path, so almost none of the conversion grant's payments go to what happens anyway. But converting gives up the running blast furnaces and brings the obligations due, so owners respond little, and the grant must reach 1098 USD per tonne of new capacity, well above the building cost, before it buys the target. It then pays about twice as much per tonne of CO_2 for added investment as the adoption grant. This comparison is particularly sensitive to the conversion response, which the model fits poorly (Section 5.2).

The conditional grant moves owners from adding an EAF and keeping the blast furnaces to adding one and later retiring them, and it pays every retirement of a hybrid that exists in 1985, including those that would happen anyway. The site-retention grant pays every tonne of EAF capacity at an old site in every year, about a third of which would have run anyway. Appendix E.1 gives this grant and the obligation takeover in detail.

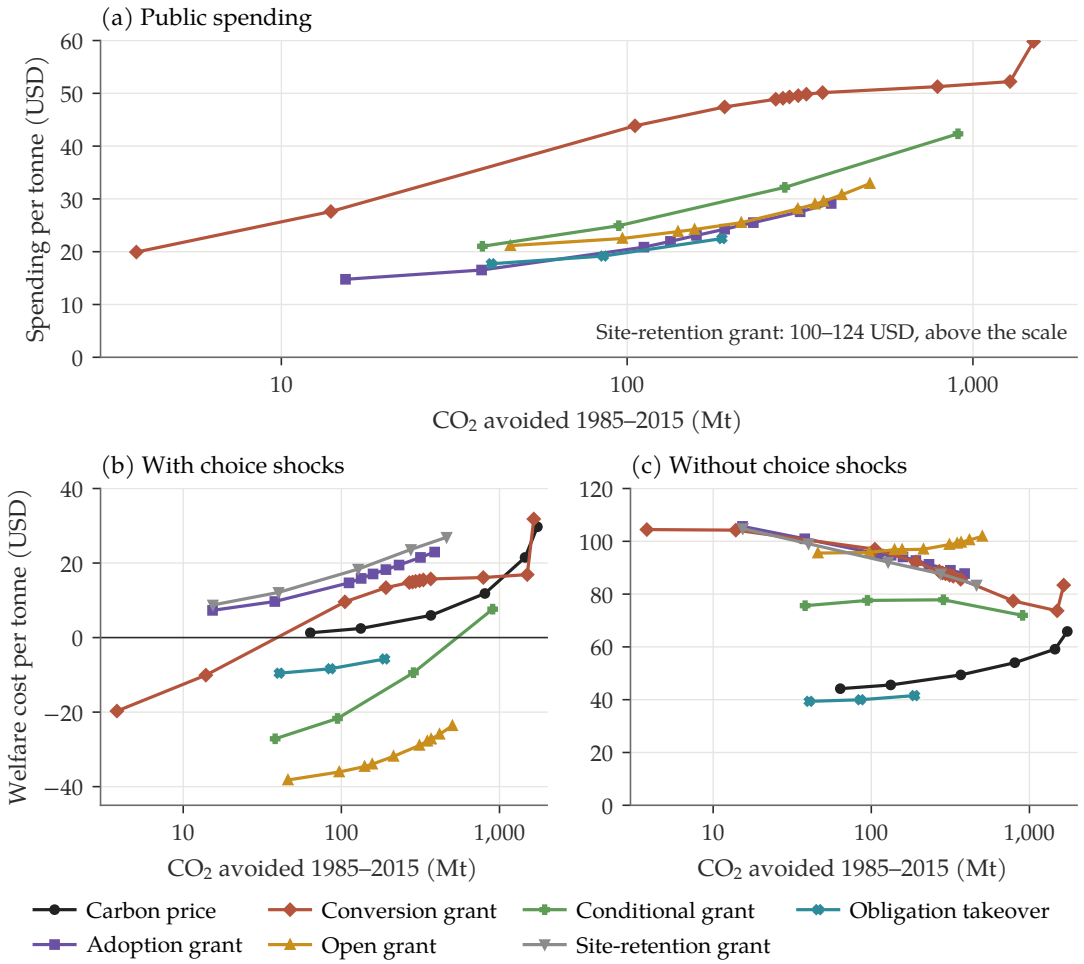
The carbon price and the obligation takeover act mainly through closure. At the target, the carbon price closes 3.4 more integrated works and adds 1.2 adoptions. The flat price rises by 24.2 USD per tonne, which draws in 5.9 new mills. Under the model's assigned obligation levels, the full obligation takeover closes 3 more integrated works and pays mainly the owners of works that close. Adoptions barely change. With entry and prices held fixed it closes more works still, but buys no output-neutral abatement.

6.3 Sensitivity of the ranking

The adoption grant's lead over the open grant is small. It disappears when the entry costs of new flat and long mills are both at the upper ends of their intervals, when steel demand responds weakly to prices, when an add-on's EAF melts enough iron from the works' own blast furnaces or when the CO_2 from EAF electricity is counted with state-by-year grid factors. Each check below re-solves the model at the abatement target unless it says otherwise. Appendix E.3 gives the details. At baseline inputs the adoption grant is the cheapest capacity grant at every target it reaches, up to 389 Mt, but its lead over the open grant shrinks as the target rises. Within its reach the obligation takeover can cost less (Figure 8).

Entry cost and scrap price. The entry cost decides how much entry comes without support, and so the base of the open grant. In the scenario in which the entry costs of both flat and long mills are at the upper ends of their 95% intervals, the open grant becomes the cheaper one, 27.5 against 29.2 USD per tonne of CO_2 (Figure 9). This is a scenario with both costs high, not a joint confidence region. With the long-mill entry cost held at its estimate, the open grant is dearer at all seven flat-mill entry costs tested across its interval, and the difference is U-shaped: at low entry costs the open grant pays for many mills that would come anyway, and at high entry costs it needs a large grant

Figure 8: Public spending and welfare cost per tonne along each instrument's curve

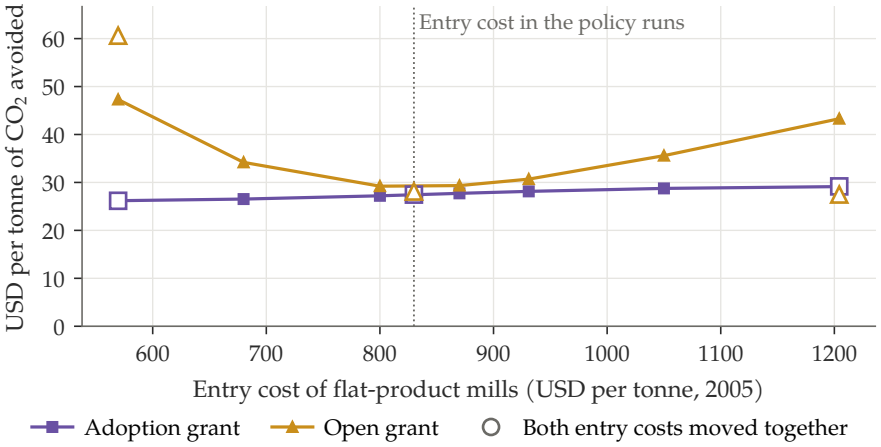


Notes: Each marker is one level of an instrument (point estimate, entry and prices responding); CO_2 avoided over 1985–2015 as in Table 4, on a logarithmic scale. (a) Grants paid over CO_2 avoided, undiscounted, USD of 2005; the carbon price raises revenue and is not shown. (b), (c) Welfare cost: the loss of steel buyers, U.S. plants and entrants and the government, Equation (17) without the value of CO_2, per tonne of CO_2 avoided, undiscounted, USD of 2005; lower is better. (b) counts in producer surplus the expected value of the logit choice shocks of plants and entrants, (c) leaves it out.

to bring mills in. The scrap price moves entry through the same channel as the entry cost. On alternative scrap price paths, at the 25th and 75th percentiles of the observed range, the adoption grant is cheaper on both: 23.4 against 36.4 USD per tonne of CO_2 when scrap is cheap, and 43.8 against 45.1 when it is dear. Cheap scrap brings more new mills that would come without support, so 78 percent of the open grant's payments go to them. These paths are alternatives, not bounds around the observed path, on which the lead is smaller than on either.

Demand and what an add-on replaces. If steel demand responds weakly to prices, the ranking also reverses. Moving the flat- and long-product elasticities together, the open grant becomes the cheaper one below a flat-product elasticity of about 0.15, against the central 0.20. That threshold lies inside the 95 percent Anderson–Rubin set of the demand estimate but below the range of 0.20 to 0.60 given for flat products in a steel

Figure 9: Spending per tonne of CO_2 and the entry cost of new mills



Notes: Each grant at the level that avoids 311 Mt of CO_2 over 1985–2015, against the entry cost of new flat-product mills (what a new mill pays per tonne beyond its steelmaking shop). Lines: the long-product entry cost held at its estimate. Hollow markers: separate runs that move both entry costs together to the low end, center and high end of their 95% intervals. Dotted line: the flat-product entry cost used in the policy runs. USD of 2005.

investigation of the U.S. International Trade Commission (USITC 2001). Input prices that move with demand may bias the estimate toward zero (Appendix C.5). The import-supply elasticity matters little. The baseline assumes that the add-on's EAF output replaces blast-furnace output. If part of the EAF's charge is iron from the works' own blast furnaces, the add-on replaces less of the dirtier route, and the adoption grant's lead disappears once that share passes a threshold. This check recounts emissions with the plants' choices held fixed.

Other checks and what is held fixed. On the joint estimate of entry and the plant model the grants cost 27.2 and 27.8 USD per tonne, with the same ranking. Across the interval for ϕ_C the result holds where the adoption grant reaches the target: at $\phi_C = 0.21$ billion it costs 23 USD per tonne and the open grant 23, and at 1.96 billion it cannot reach 311 Mt. Added to a carbon price, the adoption and open grants cost about the same per tonne of CO_2 at the two carbon prices tested. Counting the CO_2 from EAF electricity with state-by-year grid factors, instead of a fixed 0.4 tonnes per tonne of EAF steel, lowers the CO_2 each grant avoids at unchanged grant levels by 4 to 6 percent. At the abatement target the adoption and open grants then cost about the same, 29.6 and 29.5 USD per tonne. Appendix Table E.5 lists what the experiments hold fixed. Among these, steel buyers enter only as a demand curve, so their costs of switching suppliers are not counted. Counting them could favor the adoption grant if the other instruments make more buyers change suppliers. Each grant also runs on its own here, while in practice a program restricted to existing plants can run beside others that pay new plants too. The comparison does not cover such mixes.

6.4 Welfare and incidence

Spending per tonne says what a tonne of CO_2 costs the government, not whether the country gains. For that I add up the change in U.S. welfare over 1985–2015, with every flow discounted to 1985 at 2%:

$$\Delta W = \Delta \text{CS} + \Delta \text{PS} - \mathit{MCPF}\,(\text{grants} - \text{carbon revenue}) + \mathit{SCC} \cdot \Delta \text{CO}_2. \tag{17}$$

Here ΔCS is the change in steel buyers' surplus. The change in the surplus of U.S. plants and entrants, ΔPS, is operating profit, plus grants received, less carbon paid, building and entry costs, plus closure values, less cleanup and the outlays for holding, idling, restarting and relining, plus the expected value of the choice shocks of the chosen actions, $\sigma_i\big(\gamma - \log \Pr_t(a)\big)$. The shutdown obligations are payments to retirees and count as transfers, as does a fund's payment under the obligation takeover. Cleanup uses real resources. The marginal cost of public funds *MCPF* is 1.0 (an assumption), and *SCC* is the social cost of carbon. Displaced workers' earnings losses are reported as incidence, not added.

Because the instruments avoid the same CO_2 at similar dates, the term $\mathit{SCC} \cdot \Delta \text{CO}_2$ is nearly common to all of them, and differences in welfare come from the cost side. I therefore report the *break-even carbon value*, the *SCC* at which an instrument's ΔW is zero, rather than rank instruments at one assumed *SCC*, as Fowlie, Reguant, and Ryan (2016) report welfare as a function of the carbon value. One term decides the ranking: with logit entry, ΔPS counts the choice shocks of the extra entrants a grant draws in. The marginal entrant earns less than its entry cost in the model's payoffs, and its shock is what makes entry worthwhile to it. The term is consistent with the estimates, but no observed payment corresponds to it, and free entry can be socially excessive (Berry and Waldfogel 1999). I report welfare with and without the choice-shock term.

I compare the instruments at 20 abatement targets, the CO_2 avoided at each level run for the instruments (Figure 8). With the choice-shock term, the open grant has the lowest welfare cost at 19 of them. Without it, the obligation takeover has the lowest welfare cost at the 13 targets it reaches and the carbon price at the other 7. The grants never lead (Figure 8, panel c). Among the instruments that reach the target, the break-even carbon values in Table 4 are lowest for the open grant with the choice-shock term and for the carbon price without it. For reference, the value of EPA (2023) at a 2 percent discount rate is 147 USD per tonne and the interim value of the Interagency Working Group on Social Cost of Greenhouse Gases (2021) at 3 percent 39.5, for emissions in 2020 and in 2005 dollars. Welfare along other scrap paths compares policies at fixed levels, not at equal abatement (Appendix Table E.3).

At equal abatement the open grant makes steel buyers gain 30.5 billion USD and the owners of existing plants lose 41.7. The carbon price makes buyers lose 41.9 and the owners of existing plants 22.5. The adoption grant closes less existing capacity than the open grant and costs fewer jobs at existing plants, about 4,268 against 9,083. These are works jobs at existing plants only, including those lost when a works shrinks, not jobs

Table 5: The gap to DRI-EAF at the works still running in 2015

		USD per tonne of site capacity					USD bn
	Pellets	Gap	Range	Engineering	Premium	Route	Total
Scrap EAF	–	510	366–823	218	292	0	22.2
DRI-EAF, by cost of the DRI plant							
Low	Market	592	447–904	311	280	+82	25.7
Low	Captive	507	366–818	206	302	−3	22.1
Nucor	Market	690	546–1,003	428	262	+180	30.0
Nucor	Captive	605	464–916	319	286	+95	26.3
High (central)	Market	726	582–1,039	470	256	+216	31.6
High	Captive	641	499–951	361	280	+131	27.9
High, no high-end cap	Market	704	561–1,016	467	238	+194	30.6

Notes: Adoption gap (the one-time payment that would make adopting as good to the owner as the better of keeping and closing) of an EAF built beside the blast furnaces at the 13 works with blast-furnace capacity in 2015; capacity-weighted means, USD of 2005 per tonne of site capacity. DRI-EAF: the EAF melts direct-reduced iron from a plant built with it, made from iron-ore pellets with natural gas; DRI plant cost: CRU (2024) and Nucor's reported cost; pellets bought or from the owner's mines at cost. Range: the gap at the ends of the 95% interval of the fixed adoption cost ϕ_C (0.21 and 1.96 billion USD). Engineering: the engineering part, building cost and forgone profit; premium: the premium beyond engineering cost, the gap less its engineering part; route: the route premium, the gap less the scrap EAF's; total: the gap times capacity, summed over the works. No high-end cap: the DRI-EAF assumed to sell the works' own high-end share.

at new mills. They are not added to welfare and are not a net change in employment (Appendix E.5). In the data, a closure of an integrated works cut its county's employment by about 0.6 jobs per works job lost, not by a multiple (Appendix C.8).

6.5 The remaining blast furnaces

In 2015, 13 integrated works with 43.5 Mt of capacity still ran blast furnaces, and they make the grades that EAF mills reached last. The route on offer for them is an EAF fed with iron made from ore with natural gas, direct-reduced iron (DRI). Governments now pay for this route at old works and at new mills alike, such as the German aid approved for thyssenkrupp's Duisburg works (European Commission 2023) and the U.S. award for which the Middletown works in Ohio was selected (Cleveland-Cliffs 2024; DOE 2024a). At old works the support has not always been enough: ArcelorMittal dropped an approved project at its German works because even with the support it did not pay (ArcelorMittal Germany 2025), and after Cleveland-Cliffs judged that its Middletown project no longer made business sense, the U.S. Department of Energy (DOE 2026) announced an award to upgrade the works' existing blast furnace instead.

I recompute the adoption gap of Section 5.3 for these works in 2015, the last year of the model's dated path, with the EAF built beside the running blast furnaces as a gas-based DRI-EAF and the adoption parameters estimated on the scrap route carried over. The building cost adds a DRI plant to the steelmaking shop, in three cases, two from the cost estimates of CRU (2024) and one reported project, and the charge is DRI at its cash cost (Appendix E.7). I split the gap into the part an engineering comparison sees, the building cost and the forgone profit, and the rest, the *premium beyond engineering cost*. New mills on this route are not modeled.

The DRI-EAF costs more to build, not to run: the gap is 726 USD per tonne of site capacity in the central case, against 510 for the scrap EAF, and the engineering part, mostly the building cost of the DRI plant, accounts for the whole difference (Table 5). The premium beyond engineering cost is about the same as for the scrap EAF, because the obligations, the fixed adoption cost and the choice-set effect do not depend on which EAF is built. Only the saving from a smaller shop changes with the route. Owners that mine their own ore face a smaller gap: at the mines' cost of pellets the gap falls by 84 to 85 USD per tonne.

The interval for ϕ_C moves the result more than the engineering cost cases: over it, the central-case gap runs from 582 to 1039 USD per tonne. Middletown provides a comparison after the estimation period: DOE's offer of up to USD 500 million came to about 116 USD per tonne of the works' capacity, the central-case gap for the remaining works is about 6.3 times as large, and the project was abandoned, although its design, a DRI plant whose melting furnaces feed the existing oxygen furnace, differs from the model's EAF. The DRI costs come from outside the model, and hydrogen is not priced. Whom to pay depends, as in Section 6.1, on how much new capacity would come without support, how strongly each side responds and how much dirty output the new equipment replaces. If DRI-fed entry would come anyway, a grant open to new mills would again pay a large share for it; if it would not, the open grant's payments mostly buy added investment. Bachorz et al. (2026) compute the abatement that financing new direct-reduction capacity would buy. Whether that capacity would come without support is the side of the comparison this paper does not estimate.

7 Conclusion

U.S. steel became cleaner mainly through new mills, not through adoption at its old works. At baseline inputs, limiting grants to the old works saves little public spending per tonne of CO_2 avoided: it pays far less for investment that would happen anyway, but more for each tonne that the added investment avoids. The ranking reverses when the costs of setting up new mills are at the upper ends of their estimated ranges or when demand responds weakly to prices. Paying only for full conversions nearly doubles spending, a result that rests on a response the data observe rarely and the model underpredicts. In the model, the fixed adoption cost holds old works back from adopting, and the shutdown obligations steer those that adopt toward an EAF beside running blast furnaces. These are comparisons of public spending, not of welfare.

The framework applies where new clean capacity competes with existing plants and retiring an old plant brings costs due. U.S. electricity is one such case: a gas plant or a wind farm is a different plant from a coal boiler, closing a coal plant's ash units brings their closure costs due, and post-closure care where the ash stays in place (EPA 2015), and new plants account for 68% of the fall in the CO_2 intensity of generating capacity between 2001 and 2020 (Appendix E.6). Whether the ranking of grants carries over is a question for that setting's own estimates.

For the remaining blast furnaces, the U.S. history says what a government needs to know before it chooses whom to pay: how much new capacity would come without support, how strongly old works and new mills respond to grants, and how much dirty output the new equipment replaces. This paper quantifies the old works' side under 2015 costs; the side of new direct-reduction mills is not estimated. The remaining blast furnaces pose the question again: whether support should go to the old works or to new mills that may be built without it.

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A Data Construction

A.1 The plant record

The directory record. The record rests on 39 editions of the plant directories, each describing the year before (AISE 1983–2003; AIST 2004–2022). Most editions to 2014 were scanned from library copies at TIB Hannover (1983–1995), the Carnegie Library of Pittsburgh (1996–1998) and TU Bergakademie Freiberg (1999–2014); later editions are digital, from the same Carnegie Library and, for 2018 and 2020, from AIST. I read the scans with optical character recognition and extracted one record per site and year, with the owner, the steelmaking furnaces and their stated capacities, coke batteries and finishing lines. The 37 editions of 1985 to 2022 give 6,401 site-years from 1984 to 2021; the combined edition of 2001–2002 gives no record for 2001. The AIST equipment roundups, yearly furnace-level tables by process, date furnaces, check capacities and idlings, and supply output for integrated works; those from 2002 are AIST's digital editions, and I scanned the earlier ones from *Iron & Steelmaker* at the British Library.

Reconciliation with the King reports. The King capacity reports list every blast furnace, oxygen furnace (BOF) vessel, EAF and open-hearth furnace with its site, opening year, closing year, any rebuild and rated capacity in short tons (King 2025). Short tons are converted once, at presentation, at 0.90718474 metric tonnes per short ton. A site-year exists if at least one steelmaking furnace is open; the panel covers 166 sites and 4,384 site-years from 1985 to 2020. Site capacity is the sum of open furnaces; technical age is the years since the furnace opened, reset at a recorded rebuild.

A King entry is changed only when outside evidence contradicts it: two independent sources state the corrected fact, or one does and no source supports King's record. Each of the 118 applied corrections is a dated row with its sources in a correction register; they add closures King missed, re-date openings and closings, re-rate capacities, record idle spells and remove duplicate sites. A site that operated in any part of a year counts as operating, so a closure year is the first year with no operation; idle years, with capacity on the books and no output, are coded where an outside source dates the spell.

Capacity is nameplate capacity: King's rating, or the rating in the equipment roundups (ISS 1988–2003; AIST 2007–2023) or the steel plant tracker of Global Energy Monitor (GEM 2026) where these correct it. Where no rating exists, it is the furnace's heat size (the steel tapped in one batch) times a fixed number of heats a year for its type, which gives 63% of EAF capacity in 1985. King's undated EAF ratings can overstate capacity in early years, because a furnace upgraded later carries its later rating back.

Verification. Each site-year is matched to outside sources: the records of the Greenhouse Gas Reporting Program (GHGRP) and the Toxics Release Inventory (TRI) of the Environmental Protection Agency (EPA 2024; EPA 2025), the AIST directories and roundups, the plant-level cost estimates, the GEM tracker, SEC filings, the Minerals Yearbook (USGS 2025), Capital IQ's dated events (S&P Global Market Intelligence 2026a)

and a fact base of dated events with saved quotes. Sources that share an origin, such as the AIST directories and roundups, count once. Of the 166 sites, 138 are confirmed by at least two families of sources besides King, and no site in the estimation scope rests on King alone. All 19 in-scope closures that TRI can test end melting within one year of the panel's last year; the closure year of 16 in-scope sites that close before 2016, most of them small EAF mills, rests on King alone. Plant coordinates come from a geocoded list of King's sites, checked against county boundaries and facility locations in GHGRP, TRI and GEM; 52 rows were corrected.

A.2 Estimation scope

Integrated works hold a BOF. EAF mills are carbon-steel mills if any of their units makes carbon products; mills making special-quality bars count as carbon EAF mills, because they melt scrap in the same furnaces. Out of scope are 18 specialty-steel and 18 stainless sites, and 4 foundries, forges and shops that melt in induction furnaces. That leaves 26 integrated works and 100 EAF mills.

The sample has 36 closures and 30 idle years, which count as decisions to keep (Table 1). Estimation uses decisions from 1985 to 2014 and their outcomes to 2015. The decisions of 2015–2019, with their outcomes in 2016–2020, are held out: 434 decision years at 88 sites, with 3 closures and 1 conversion, Fairfield's.

The estimation sample has 8 closures of integrated works and hybrids in its decisions of 1985–2014 (6 integrated works and 2 legacy hybrids). Table 1 counts 28 closures of EAF mills, which include the EAF steelmaking shops of the Farrell works and of U.S. Steel's South Works, whose BOF had closed before 1985. The likelihood counts 29 and has 2,760 decisions against the table's 2,763, because the timing states recode two sites: Kentucky Electric Steel closes in its last operating year and its later idle years are dropped, and the construction year of Lorain's conversion is dropped.

King lists the Farrell works of Sharon Steel as two sites, one with its oxygen furnaces and one with its EAFs, which close together when the works shut down in November 1992; the sample counts one corporate closure as an integrated and an EAF closure. Two other sets of same-year closures are single corporate events: Birmingham Steel's shutdown of two steelmaking shops and consolidation of a third (fiscal 1991), and RG Steel's idling of Sparrows Point and Warren after its Chapter 11 filing in May 2012. The model treats each closure as a separate decision.

A.3 Owners and obligations

Ownership. The owner of each site-year is the company heading the site's entry in the AIST directory of the following year's edition. Where no edition has the site, the owner comes from the GHGRP parent company, the fact base or the plant cost model's owner history, in that order. A firm is the controlling group on 31 December (an equity link above one half); a 50/50 joint venture is a firm of its own. Changes of owner are dated by completed deals from SDC and Orbis (London Stock Exchange Group 2026; Moody's

Analytics 2026), which override the lag of the directories. The GHGRP parent agrees with the table in 99.7% of the site-years both cover; King's own owner codes agree in only 76% of site-years before 2002 and 93% after. Of the 174 changes of owner or firm, 156 are dated by a completed deal.

Obligations. For each firm and year the table holds the pension plans' projected benefit obligation and assets, the retiree health obligation and the date on which the PBGC became trustee of the firm's plans. Pension values come from Compustat's pension file (S&P Global Market Intelligence 2026b), replaced where possible by the notes to the firm's 10-K filings (18 filers, fiscal years to 2005), and the projected benefit obligation is identical in 123 of the 125 firm-years that both cover. Retiree health obligations come from the 10-K notes, with Compustat's net balance-sheet amount used where no note is saved. The dates on which the PBGC took over the plans of 22 steel firms, 1983–2010, come from the PBGC's list of trusteed plans; from that date the plan's unfunded obligation is the PBGC's. Records are matched to plants through the owner table, and a firm's latest record is carried forward for at most three years. The obligation per tonne divides the unfunded obligation, pension obligation less plan assets plus the retiree health obligation, by the firm's steelmaking capacity in the panel. The model sorts firm-years into three levels (none, partial, full) by this obligation per tonne. Matched to owners, the records cover 53% of the sample's capacity.

Cleanup accruals. Cleanup accruals come from the saved annual reports of the integrated firms, fiscal years 1993 on; Table A.1 lists them by owner. Readers label each passage with an environmental accrual or reserve as a firm total, a site-specific amount, an asset retirement obligation or not usable, and every quoted amount is checked against the filing. The result is 107 owner-years of firm totals. A site-specific amount is used where the filing names the works; otherwise the firm total is split by BOF capacity, which gives a value for 268 works-years in 2005 USD. Asset retirement obligations are not added. Works-years without a record take the mean.

A.4 The three adoptions

Table A.2 gives the three adoptions of the EAF by integrated works between 1985 and 2020.

B Prices, Costs and Margins

All prices and costs are in 2005 U.S. dollars per metric tonne, deflated with the GDP deflator (BEA 2026a).

Table A.1: Site cleanup accruals of the integrated firms

Owner	Years	Firm-years	Accrual (million USD)		
			Median	Maximum	Per tonne
International Steel Group	2003–2004	2	216	221	10.9
U.S. Steel	1993–2015	23	150	201	11.2
LTV	1994–2000	7	131	181	12.2
Bethlehem	2002	1	125	125	14.5
AK Steel	1993–2015	23	53	111	10.4
Ispat Inland	1998–2004	7	30	39	5.7
Inland	1993–1997	5	29	32	5.4
Republic	1999–2002	3	22	25	1.1
National	1993–2002	10	14	28	2.0
Wheeling-Pittsburgh	1993–2006	14	13	21	4.3
Weirton	1996–2002	7	10	10	2.3
Acme	1993–1997	5	0	1	0.7

Notes: Environmental cleanup accruals and reserves in the owners' annual reports (10-K), fiscal years 1993 on, million USD of 2005. Firm-years: years with a usable accrual. Per tonne: median over works-years of the accrual per tonne of the owner's BOF capacity, site-specific where the filing names the works, else the firm total split by BOF capacity. Accruals are lower bounds: firms book only what is probable and estimable, at the low end of a range. BOF: basic oxygen furnace.

B.1 Steel prices and product mix

Prices. The Current Industrial Reports on steel mill products (U.S. Census Bureau 1983–2010) give each product's quantity shipped and its value at the plant gate, so their ratio is a realized price. I extracted all 27 editions from 1983 to 2010 and checked each of the 7,095 rows against its source line. The hot-rolled and cold-rolled lines are used to 2001. From 2002 the flat prices are the hot-rolled band and cold-rolled coil prices of SteelBenchmarker (2026) times the median ratio of the Census value to them over 2002–2010 (0.896 and 0.971). After 2010, coated sheet follows the import unit value of galvanized sheet (times 0.905) and rebar the producer price index for hot-rolled bars, plates and shapes (times 3.288), each on the Census level.

Plant prices. Each in-scope site has, for each year, shares of its output across the flat products (hot-rolled, cold-rolled and coated sheet, tin-mill products, plate) or the long products (rebar, merchant bar, shapes, wire rod). A prior from the finishing lines that the AIST directories print for the works is calibrated to the product shipments that firms report in their 10-K filings or Compustat product segments; from 2006 the plant-level sales estimates replace it. The plant price is the share-weighted mean of the products' Census unit values. Relative to firms' reported revenue per tonne shipped, the base-product price gives 0.75 on average and the product mix 0.94.

Finishing costs. A product beyond the base product (hot-rolled sheet for flat works, rebar for long works) also costs more to make. Its finishing cost is its price premium over the base product less its margin premium in the product-line accounts of the U.S. International Trade Commission (USITC 1993–2017), from its reports in steel investigations, so that the recipe reproduces the margin differences producers reported. Before

Table A.2: The three adoptions

	Steubenville	Lorain	Fairfield
Type	Add-on	Conversion	Conversion
Products after adoption	Flat	Long	Long (rounds)
BOF capacity before (Mt)	3.46	2.54	2.25
EAF capacity after (Mt)	2.42	1.18	1.51
EAF over old capacity	0.70	0.46	0.67
Owner bankruptcy before	2000	2001, 2003	None
Blast furnaces stopped	2008 [a]	2008	2015
EAF first heat	2004	2013	2020
Decision year in the model	2003	2011	Holdout

Notes: Add-on: an EAF built beside running blast furnaces; conversion: the blast furnaces shut and an EAF built in their place. Capacity in Mt a year from the plant panel; Steubenville's BOF capacity after the add-on is 1.73 Mt. EAF over old capacity: EAF capacity after over integrated capacity before. Decision year: the decision the model reads as the adoption; holdout: after the estimation years. [a] The year the oxygen-furnace shop was idled, as reported in the 2012 BOF roundup of the Association for Iron & Steel Technology (AIST); King's capacity reports retire the blast furnaces in 2010. EAF: electric arc furnace; BOF: basic oxygen furnace.

1990 the USITC level of 1990–92 is carried back.

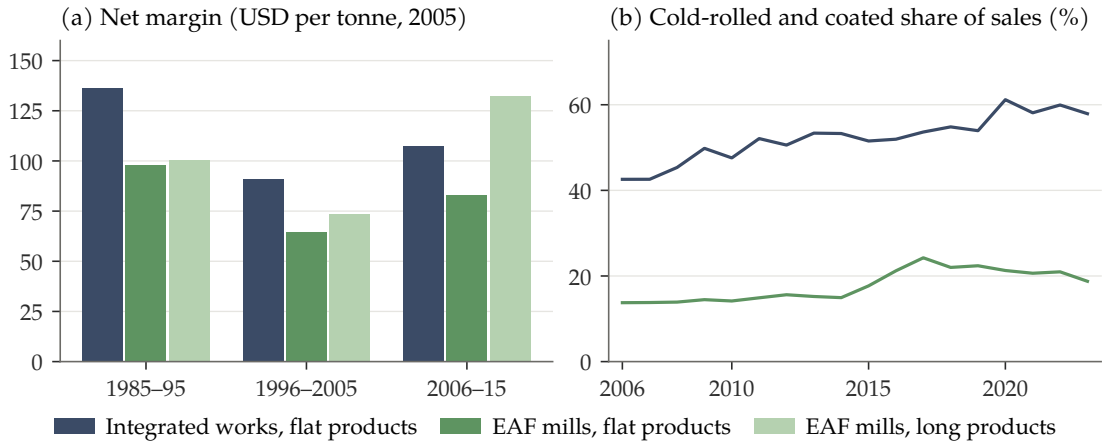
The cap on the new EAF's high-end products. After an adoption the new EAF sells no tin plate, and its cold-rolled and coated share is capped at the share that EAF sheet mills reached, $\bar{w}_t = 0.32$, 0.30 and 0.21 in 1985–95, 1996–2005 and 2006–15; the excess is sold as hot-rolled sheet. Figure B.1b shows the high-end shares of integrated works and EAF flat mills from 2006; the cap uses EAF sheet mills, weighted by output, and so lies above the line for all EAF flat mills, which include plate mills.

B.2 Costs

Input prices and coefficients. The cost recipe takes its coefficients and its levels from 2006 on from the cost estimates of CRU (2024). Iron ore is the average value at the mine (USGS 2025) times 1.147, which scales it to Cleveland-Cliffs' realized pellet price, and coking coal the price of coal delivered to coke plants (EIA 2026b). Scrap prices are the published prices of the main U.S. scrap markets, and electricity prices the industrial prices of the plant's state (EIA 2025c). EAF metallics cost 1.18 times the scrap price for flat products from 1989, more before (the quality penalty of Section 4.1), and 1.10 times for long products.

Non-material cost. The cost estimates give labor, overhead and other operating cost by route only from 2006. Hours per tonne and other costs are set to the levels of Barnett and Crandall (1986) for representative plants of 1985 and to the cost estimates in 2006–2015; hours per tonne of raw steel set the timing in between, and labor cost is hours per tonne times compensation per hour (BLS 2026a), including benefits, times the route's relative wage. Casting and rolling costs the route's mean in the cost estimates, with a premium for casting in ingots at integrated works that vanishes by 2000; for rebar the level is calibrated to USITC's cash cost less the recipe's liquid-steel cost over 1996–2016.

Figure B.1: Net margins by route and the high-end share of sales



Notes: (a) Net margin, USD of 2005 per tonne: the plant price at its product mix less material, energy, labor, other non-material and finishing costs, before capital costs; output-weighted across plants within each year, then averaged over the years of each period. Flat products: sheet and plate; long products: bars, rods and sections. (b) Cold-rolled and coated products as a share of finished sales, capacity-weighted, 2006–2023. EAF: electric arc furnace.

Building cost. The building cost of an adoption is the replacement cost of an EAF steelmaking shop in the cost estimates, 260 USD per tonne of EAF capacity (median over U.S. EAF mills, 2020). It excludes rolling and finishing mills, which an adopting works already has.

B.3 Margins against reported profits

The model's net margin is the plant price at its own product mix minus material, energy, labor, other non-material and finishing costs, before capital costs. At integrated flat works it averaged 136 USD per tonne in 1985–95, 91 in 1996–2005 and 108 in 2006–15; at EAF flat mills 98, 64 and 83. The margins leave out selling and administrative expense, retiree pension and health expense, restructuring charges and the cost of idle capacity, so they are not reported profits (Figure B.1).

Table B.1 compares the recipe with the product-line accounts of USITC. The rebar cost level is calibrated to them, so its cash cost matches on average by construction; its year-to-year movement is a test, and the recipe follows it but misses the swings of 2007–2009 and 2015–2016. The hot-rolled margin is not calibrated: it is within 30 USD per tonne of USITC's in each decade and moves with it, except in 1990–92 (annual data, not shown), when the recipe lacks the fixed costs that rise per tonne at low utilization.

Reported profit is operating income before depreciation per tonne shipped, from 10-K filings and Compustat; the comparable basis adds back the expenses the margins leave out, except idle capacity. For 1985–95, the integrated advantage over EAF long producers on the comparable basis is 92 against 75, half the model's 136 against 100: the sign holds and the size is overstated. After 1995 the reported ordering is reversed for the 2 listed EAF flat producers (126 and 139 against 41 and 61 for integrated firms), but those firms sell plate, structural shapes and rail and are not representative of sheet mills.

Table B.1: Recipe against USITC product-line accounts

			Price		Cash cost		Margin	
Product	Years	Obs.	USITC	Recipe	USITC	Recipe	USITC	Recipe
Hot-rolled	1990–1999	7	433	441	396	380	37	61
Hot-rolled	2000–2009	10	509	512	440	413	69	99
Hot-rolled	2010–2015	4	571	553	529	521	42	33
Rebar	1996–1999	4	382	389	328	340	53	48
Rebar	2000–2009	8	455	442	375	375	80	68
Rebar	2010–2016	7	565	559	500	494	64	65
Cold-rolled	1990–1992	3	627	654	571	460	57	194
Cold-rolled	2013–2015	3	670	627	633	576	36	51

Notes: USD of 2005 per tonne; means over the years with data in the product-line accounts of the U.S. International Trade Commission (USITC), first and last year shown, some spans with gaps. Obs.: years with data. USITC: net sales, cost of goods sold less depreciation, and their difference. Recipe: the model's price, the cash cost of liquid steel plus casting and rolling, and their difference; hot-rolled sheet is the capacity-weighted mean of the two routes. The rebar level is calibrated to these data; only its time pattern is a test.

For sheet mills in 2006–15, the cost estimates put the model's EAF margin within 8 USD per tonne of theirs (71 against 78) and keep the ordering. The integrated advantage in the model's margins is therefore not a statement about reported profits; the obligations and the idle-capacity costs enter the adoption comparison separately.

B.4 The price process

The plant model needs expectations of future margins, which three margin indices summarize: the national price of hot-rolled sheet or rebar minus the material and energy cost of the route (integrated flat, EAF flat, EAF long), averaged over the route's in-scope plants with capacity weights. Each index follows a stationary autoregression of order one with correlated innovations, estimated as a system of seemingly unrelated regressions on 1986–2015 (Table B.2). No index has a significant trend: with a common flat persistence the trends are -0.9 (standard error 1.8), -2.0 (1.2) and -0.2 (1.0) USD per tonne a year. The persistence is 0.25, 0.44 and 0.48 for the integrated flat, EAF flat and EAF long indices. A common flat persistence (0.38) is rejected ($p = 0.009$), and the Bayesian information criterion prefers own persistence, 957.3 against 961.1. Demand shifters, import shares and the aggregate EAF share worsen the holdout forecasts, so no aggregate state beyond the margins enters the model. Dynamic forecasts of the 2016–2020 margins, which were not used in estimation, have a root mean squared error of 57 USD per tonne, against 105 for a random walk from 2015 and 65 for the specification with trends.

C Descriptive and Reduced-Form Evidence

C.1 Decomposition variants

The decomposition of Section 3.1 splits the change in the capacity-weighted CO_2 intensity $\bar{e}_t = \sum_i s_{it} e_{it}$ of the 166 plants of the panel, where s_{it} is plant i's share of fleet

Table B.2: The price process of the margin indices, 1986–2015

	Constant	Trend	Persistence	Mean	Innovation sd	RMSE	BIC
A. Own persistence							
Integrated, flat	253 (43)	–	0.25 (0.12)	338	–	71	
EAF, flat	139 (30)	–	0.44 (0.11)	248	–	60	
EAF, long	117 (30)	–	0.48 (0.13)	223	–	31	
All indices						57	957.3
B. Common flat persistence							
Integrated, flat	209 (39)	–	0.38 (0.10)	337	90	72	
EAF, flat	155 (29)	–	0.38 (0.10)	249	61	58	
EAF, long	115 (30)	–	0.49 (0.13)	223	44	31	
All indices						56	961.1
C. Common flat persistence and trends							
Integrated, flat	275 (51)	−0.9 (1.8)	0.23 (0.12)	–	86	72	
EAF, flat	224 (41)	−2.0 (1.2)	0.23 (0.12)	–	57	79	
EAF, long	127 (39)	−0.2 (1.0)	0.44 (0.14)	–	44	33	
All indices						65	963.4
Random walk from 2015						105	

Notes: USD of 2005 per tonne. Margin index: the national price of hot-rolled sheet or rebar minus the material and energy cost of the route, averaged over the route's in-scope plants with capacity weights. Each index regressed on its lag, and in panel C on a trend, as a system of seemingly unrelated regressions on 30 transitions; standard errors in parentheses. Trend: USD per tonne a year. Mean: the constant over one minus the persistence. Innovation sd: the standard deviation of the innovation, not computed for panel A. RMSE: root mean squared error of dynamic forecasts of the held-out years 2016–2020 from the 2015 margins. BIC: Bayesian information criterion of the system. EAF: electric arc furnace.

capacity and e_{it} its emission intensity in tonnes of CO_2 per tonne of capacity: 2.0 for blast-furnace capacity, and for EAF capacity 0.08 of direct emissions plus 0.52 MWh of electricity per tonne times the CO_2 factor of the state's power sector in that year (EIA 2026c; EIA 2025c). Following Olley and Pakes (1996), extended to entry and exit, with subscripts 0 and 1 for 1985 and 2020 and *Entry*, *Exit* and *Stay* the plants that open, close or operate in both years,

$$\Delta\bar{e} = \underbrace{\sum_{i\in Entry} s_{i1}(e_{i1}-\bar{e}_0)}_{\text{entry}} - \underbrace{\sum_{i\in Exit} s_{i0}(e_{i0}-\bar{e}_0)}_{\text{closures}} + \underbrace{\sum_{i\in Stay} s_{i0}\,\Delta e_i}_{\text{change at surviving sites}} + \underbrace{\sum_{i\in Stay} \Delta s_i(e_{i1}-\bar{e}_0)}_{\text{reallocation}}. \tag{18}$$

The technique channel of Shapiro and Walker (2018) is shut off by construction. Table C.1 compares the headline split with a fixed EAF intensity of 0.4 and with variants that move the direct emissions and the electricity use, hold the grid at its 1985 factors, or price net electricity imports at the U.S. factor. Holding the grid at its 1985 factors for all plants, new mills included, cleaner electricity accounts for about 26% of the decline (23 to 31 percent across the parameter range); the cleaner-power channel of Table C.1 counts only EAF capacity that operates in both years; entry is the largest channel in every variant.

Table C.1: Decomposition of the fall in CO_2 intensity under alternative EAF intensities

	Fixed	Grid	Grid 1985	Low	High	Consumption
EAF parameters	0.4	0.08, 0.52	–	0.04, 0.45	0.15, 0.60	–
Intensity 1985	1.247	1.279	1.279	1.236	1.340	1.280
Intensity 2020	0.947	0.886	0.989	0.839	0.955	0.882
Fall (%)	24.1	30.7	22.7	32.1	28.7	31.1
Share of the fall (%)						
Entry	88.1	75.4	87.2	77.2	73.1	75.2
Surviving plants	19.1	29.0	18.3	27.6	30.8	29.1
Route change	–	13.5	18.3	14.3	12.5	13.4
Cleaner power	–	15.5	0.0	13.3	18.3	15.6
Closures	−11.4	−8.9	−12.0	−9.2	−8.4	−8.6
Integrated works	33.4	24.5	33.2	25.7	22.9	24.1
EAF mills	−44.8	−33.4	−45.2	−35.0	−31.3	−32.7
Reallocation	4.2	4.5	6.5	4.5	4.5	4.4

Notes: Capacity-weighted CO_2 intensity of all plants in the panel, tonnes per tonne of capacity, and shares of its 1985–2020 fall by channel, Equation (18). Variants differ in EAF intensity: fixed; grid-based (the baseline of Figure 2), direct emissions plus electricity times the state's power-sector CO_2 factor (EAF parameters: direct emissions in tonnes of CO_2 and electricity in MWh per tonne; for the fixed variant, the intensity); low and high: the grid variant with low and high parameters; grid at 1985 factors; consumption-based, with net electricity imports at the U.S. factor. Route change: adoption, and EAF steelmaking shops closed at surviving works; cleaner power: the fall in grid factors of surviving EAF capacity; reallocation: shifts in capacity among surviving plants. –: not reported.

C.2 What integrated firms said and bought

Outside their own works, integrated firms adopted EAF capacity in 14 cases over 1985–2021: 2 ventures building new mills, 6 targeted purchases and 6 EAF mills bundled in whole-company deals. The firms free of the pension plans after a PBGC takeover, Wheeling-Pittsburgh, whose plans the PBGC took over, and ISG and Severstal, which bought works without the plans, acted within a few years. The firms that still carried theirs (LTV, Bethlehem, AK Steel, U.S. Steel) bought plate, specialty-steel and tubular businesses, or took part in LTV's Trico flat mill, which closed within a few years, and bought flat EAF capacity only from 2019.

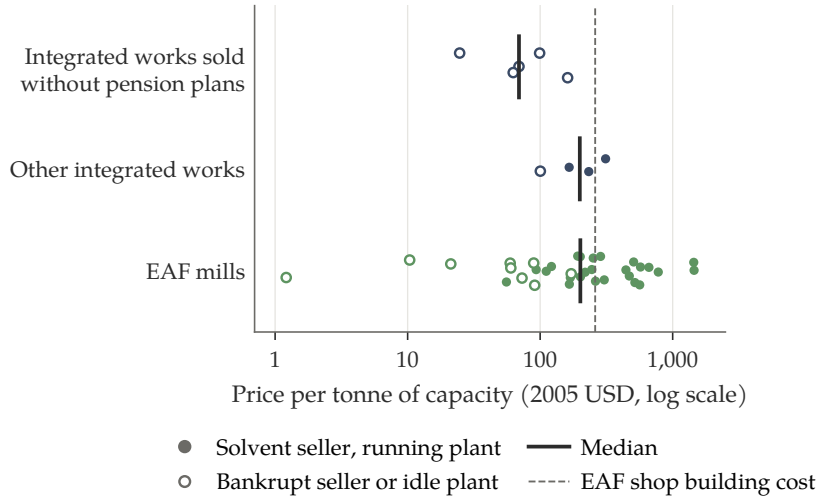
C.3 Deal prices

Figure C.1 shows what buyers paid for a going works once the obligations stayed behind. The median over 10 integrated deals with a usable value is 119 USD (2005) per tonne of steelmaking capacity, less than half the cost of an EAF steelmaking shop. The 5 works bought out of bankruptcy without the seller's pension plans sold for a median of 69, Inland, bought with its plans in 1998, for 312, and EAF mills for 201.

C.4 Reline or close, and the distress test

A blast furnace must be relined, its refractory lining rebuilt, every one to two decades, so the choice to reline or close falls due at an age that engineering largely sets. The risk set is every furnace-year at the 24 integrated works of the estimation sample in 1986–2015 in which the furnace is of unknown age or at least 11 years into its campaign,

Figure C.1: What buyers paid for steel plants, 1985–2020



Notes: Carbon-steel plant sales with a usable value, 1985–2020, USD of 2005 per tonne of the target's steelmaking capacity, log scale; one dot per deal, placed at random heights within its row; the sale of integrated works together with an EAF mill (2011) is not shown. Integrated works sold without pension plans: bought out of bankruptcy without the seller's plans. Hollow: the seller was in bankruptcy or the plant idle at the sale, from court, company and press records. Dashed line: the building cost of an EAF steelmaking shop, 260 USD per tonne. EAF: electric arc furnace.

the age by which a quarter of campaigns with a known start have ended in a reline: 991 furnace-years of 72 furnaces, with 45 relines, 33 closures and 2 bankings (furnaces taken out of service for a time). Obligations are the owner's pension and retiree health obligations per tonne of its capacity, gross of plan assets, not the shutdown costs the model charges. Regressors are dated $t-1$, and logits are penalized (Firth 1993). The design assumes that, once a reline is due, its timing does not respond to unobserved shocks that also move the regressors; this holds only in part, since a works in decline stops relining.

Distress could change closure decisions and the obligations an owner carries at the same time: owners in Chapter 11 carried 293 USD per tonne against 496 for owners that had never filed. Table C.2 therefore adds distress measures dated $t-1$ to the logits of closing against continuing (944 furnace-years, 33 closures) and against relining (78 resolutions): Chapter 11, earlier filings, leverage, operating income over assets, negative book equity, and the years after a PBGC takeover.

Against continuing, the sign of obligations survives distress: the odds ratio is 0.73 alone (95% interval 0.59–0.88), 0.80 with the Chapter 11 state, 0.82 with the balance sheet and works controls (0.65–1.00, the interval just reaching one), and 0.70 within owners over time. Distress is the larger predictor: Chapter 11 multiplies the odds of closing by 4.22 (2.00–8.82). Closure rates jump at the filing, from 2.2% a furnace-year three or more years before it to 8.9% in the filing year and the next. A takeover with no Chapter 11 in force, the one cell that would isolate the removal of obligations from distress, has 120 furnace-years and 2 closures, too few to detect an effect. The association is consistent with obligations delaying closure, and Chapter 11 timing may itself anticipate the shutdown costs, which the controls cannot absorb.

Table C.2: Obligations and Chapter 11: odds ratios of closing a due blast furnace

	Close vs continue		Close vs reline	
Specification	Obligations	Chapter 11	Obligations	Chapter 11
Obligations alone	0.73 [0.59, 0.88]	–	0.78 [0.61, 0.97]	–
+ Chapter 11 state	0.80 [0.65, 0.96]	4.22 [2.00, 8.82]	0.82 [0.63, 1.03]	3.54 [1.21, 11.17]
+ balance sheet and works controls	0.82 [0.65, 1.00]	6.42 [2.23, 18.65]	0.75 [0.51, 1.01]	17.64 [1.89, 421.45]
+ post-takeover indicator	0.80 [0.65, 0.97]	4.04 [1.86, 8.70]	0.86 [0.66, 1.08]	3.53 [1.20, 11.25]
Within owner (deviation from owner mean)	0.70 [0.53, 0.91]	4.56 [2.10, 10.08]	0.68 [0.39, 1.01]	4.09 [1.36, 13.44]
Observations	944 furnace-years		78 resolutions	
Closures	33		33	

Notes: Odds ratios of closing a blast furnace due for relining (of unknown age, or at least 11 years into its campaign), penalized (Firth) logits, 1986–2015; 95% profile-likelihood intervals in brackets. Close vs continue: over furnace-years; close vs reline: over the years a due furnace was relined or closed. Obligations: the owner's gross pension and retiree health obligations, per 100 USD per tonne of capacity. Chapter 11: owner in bankruptcy reorganization. Rows add the Chapter 11 state; balance-sheet and works controls (on the 829 furnace-years and 72 resolutions with every regressor); to row 2, the years after a PBGC takeover of the pension plans; last row, obligations as deviations from the owner's mean. Regressors dated $t-1$. PBGC: Pension Benefit Guaranty Corporation.

C.5 Demand elasticities

The counterfactuals of Section 6 let steel prices respond to policy. I estimate the price elasticity of demand from annual Census shipments by product, 1985–2010, with product trends and demand shifters, instrumenting the price with the material cost of the producing route. Where the instrument is strong the short-run demand curve is close to vertical: -0.03 pooled (95% Anderson–Rubin set $[-0.17, 0.10]$), 0.03 for flat and -0.10 for long products (Table C.3). Input prices still move with demand, which may bias these estimates toward zero.

The counterfactuals use 0.20 for flat and 0.25 for long products as central values of the absolute elasticity, both more elastic than the point estimates. The long value lies inside the long set ($[-0.28, 0.10]$) and at the low end of the range of 0.25 to 0.75 given for rebar by the U.S. International Trade Commission (USITC 2017); the flat value lies just beyond the flat set ($[-0.18, 0.21]$) and at the low end of the range of 0.20 to 0.60 given in an earlier steel investigation (USITC 2001). The sensitivity runs of Appendix E.3 use 0.05 and 0.40 for flat and 0.10 and 0.60 for long products.

The elasticity of import supply comes from annual changes in log imports on changes in the log U.S. price, 1994–2015. OLS gives 0.70 for flat and 0.68 for long products. Instrumenting the price with the growth of U.S. durable-manufacturing output, which shifts U.S. demand and not foreign supply, gives the central values 1.48 and 1.30, with upper interval ends 3.17 and 2.19. The instrument is weak for flat products and invalid if the world steel cycle moves with U.S. durable manufacturing. The sensitivity runs use the OLS estimates as low and the upper interval ends as high values.

Table C.3: Price elasticity of domestic steel shipments, 1985–2010

	Elasticity	95% Anderson–Rubin set	First-stage F
Pooled	−0.03	[−0.17, 0.10]	173
Flat products	0.03	[−0.18, 0.21]	134
Long products	−0.10	[−0.28, 0.10]	147

Notes: Price elasticity of domestic shipments of steel mill products, by two-stage least squares on annual product data, 1985–2010, with product trends and demand shifters; instrument: the material cost of the producing route. Anderson–Rubin set: 95% confidence set robust to weak instruments; first-stage F: F statistic of the excluded instrument.

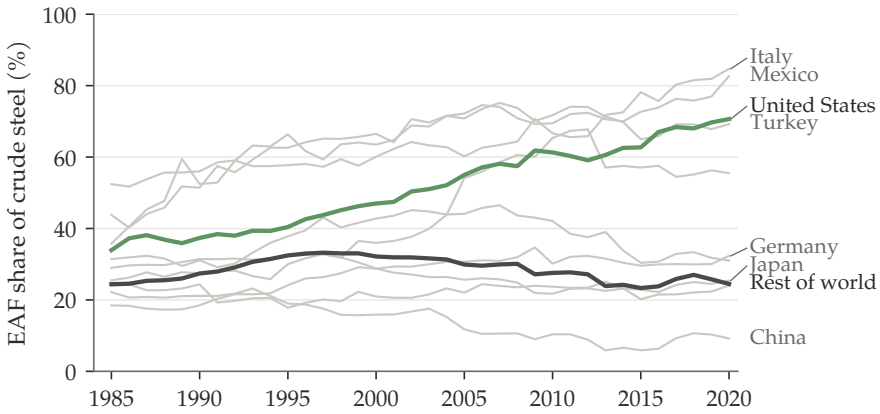
C.6 Other explanations of the shift

Demand and imports. Added one at a time to the margin process (Appendix B.4), only durable-manufacturing growth improves the fit, and it matters for EAF long margins (3.6 USD per tonne per point of growth, standard error 1.5) more than for integrated flat margins (3.4, 2.7). Demand moves the margins of both routes together; the model carries it through the margin process. One point more of last year's import share goes with an integrated flat margin −10.1 USD per tonne lower (standard error 4.8), against −5.6 (3.4) for EAF flat; broad trade measures are imprecise (9.5, 43.4). Imports enter the model through the margins of integrated works.

Energy prices. After 2008 the real gas price fell 73%, but electricity at EAF flat and long plants fell only 7% and 11%; electricity explains 1.9 of the −263 USD change in the EAF flat margin index over those years, against −445 for the steel price. Cheap gas mattered for differences in EAF costs across states more than for the industry margin, and it came late in the period.

Regulation. Air-quality rules bind through nonattainment designations, which tighten permitting in a county; EPA's records give county status only from 1992. Integrated works sat in nonattainment counties in almost every year, EAF mills far less often, and in-scope sites closed at 1.7 % a year in nonattainment counties against 0.8 % elsewhere, an association confounded with technology, age and firm. The two national coke-oven rules of 1993 and 2003 forced a rebuild-or-close decision on works with their own coke batteries; works with a battery did not close more after the rules than works buying coke (−1.7 points a year, −5.0 to 1.6), a null with wide intervals rather than evidence that the rules did not matter. Integrated works expected to face more enforcement were more likely to close (Deily and Gray 1991); across manufacturing, nonattainment slowed the growth of polluting plants relative to the same industries in attainment counties, for iron and steel clearly so under the carbon monoxide rules (Greenstone 2002), and rules stricter for new sources and major modifications give owners a reason to keep old plants running (Stavins 2006; Bialek-Gregory et al. 2025). Regulation added to the pressure on old integrated works; the data do not show it as the force that decided between converting and closing.

Figure C.2: EAF share of crude steel by country, 1985–2020



Notes: Share of crude steel made in electric arc furnaces (EAF), in percent, 1985–2020. Rest of world: world production less the United States. Unlabelled lines: other large producers.

C.7 The scrap price and the EAF share abroad

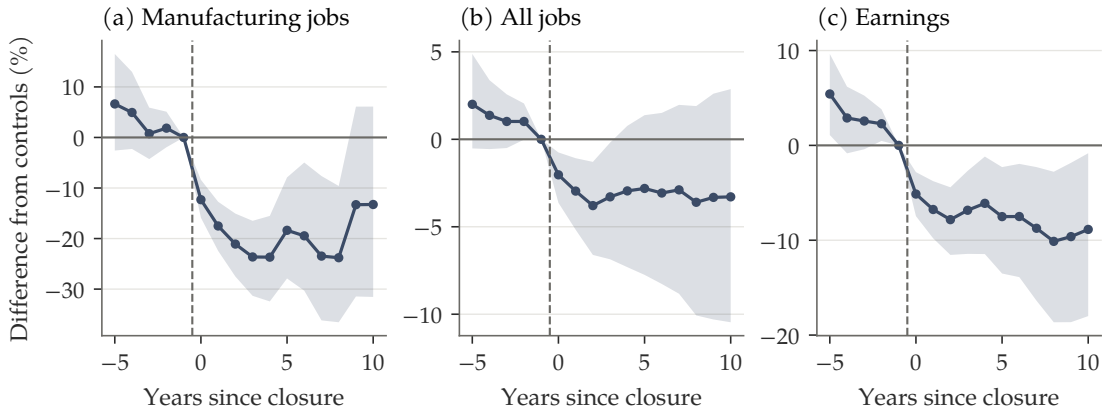
Figure C.2 shows the EAF share of crude steel by country: the rest of the world stood at 24% in 1985 and 24% in 2020, while Italy and Mexico reached higher EAF shares than the United States, and Turkey about the same. Regressing the log U.S. scrap price on the log world price and the U.S. EAF share, 1989–2015, gives 1.2 % per point of EAF share and 0.85 % with a time trend, so feedback cannot be told apart from a trend.

C.8 Local effects of closures

A conversion keeps part of a site alive; a closure does not. I compare each county with a closure of an integrated works, 8 events in 1993–2016, with its 20 nearest counties in the same Census region that host no steel-plant event within ten years, matched on population, the manufacturing and steel shares of jobs, income and employment growth, with county jobs and earnings from the Bureau of Economic Analysis (BEA 2026b) and the Quarterly Census of Employment and Wages (BLS 2026b). The estimator is a stacked difference in differences relative to the last full operating year, with inference by randomization over the matched controls; it assumes that, absent the closure, the treated county would have moved with its controls. Matching cannot find counties as dependent on steel as the treated ones.

A closure took away the works' manufacturing jobs and the earnings they paid, and little more that can be detected (Figure C.3). Earnings were 3.3 percent above the controls' path in the pre-period, so part of the decline predates the closure, as the works shrank; removing each event's pre-trend leaves impact-year effects of -1.7 percent for employment and -4.0 percent for earnings. Per steel-mill job lost, county employment fell by about 0.6 jobs, not a multiple, and earnings at county workplaces by about USD 57 thousand a year.

Figure C.3: County outcomes around closures of integrated works



Notes: 8 closures of integrated works, 1993–2016. Each treated county minus its 20 matched control counties, relative to the year before the closure, log points times 100; shaded: 90% randomization intervals; dashed line: the closure. Manufacturing jobs from the Quarterly Census of Employment and Wages; all jobs and earnings by place of work from the Bureau of Economic Analysis.

D Model and Estimation

D.1 States and payoffs

Configurations and nodes. A configuration is its BOF and EAF capacity as shares of the spell's site capacity K. An integrated works can reach the add-on hybrid H^{add} (EAF share κ_{add}), the EAF mill E^{conv} after a conversion (share κ_{conv}) and the EAF mill E^{ret} after the hybrid retires its blast furnaces; a legacy hybrid i keeps its observed EAF share κ_i^{leg}. Every configuration has an idle node. A node also holds the campaign age of the main blast furnace, up to a cap of 25 years; idle years do not age it and a reline resets it. A conversion passes through $T^{\text{conv}} - 1$ construction years with building-cost installments and no steel, an add-on through $T^{\text{add}} - 1$ years while the blast furnaces run. Capacity otherwise follows the data; a change in the data starts a new spell that the owner had not expected.

Prices and time. The three margin indices follow the price process of Appendix B.4: a joint chain of all three for flat sites (Tauchen, 125 states), their projection for the two long integrated works, and a 7-state Rouwenhorst chain for EAF long mills. After 2015 costs are frozen at their 2015 values, utilization is its 2011–15 mean and the furnace-replacement hazard is 0.28 per 100 mill-years.

Obligation levels. A firm's obligation is its unfunded pension obligation (the projected benefit obligation net of plan assets) plus its retiree health obligation net of any assets, per tonne of the firm's capacity, from the latest record no more than 3 years before the decision. The level is $L_f = 0$ below 5 USD per tonne, $L_f = 1$ at or above 50, and 0.5 in between or when no usable record exists and is treated as permanent. The scale $\bar{\lambda}$ is the mean obligation per tonne at $L_f = 1$ over the decisions of integrated works. On the gross basis the scale would be 520 USD per tonne (Section 5.4).

Table D.1: Calibrated objects

Object	Value	Source
Utilization U_t^r, U_i^{p}	National rate by route and year; plant factor	Production by route; plant output estimates, Toxics Release Inventory, AIST roundups
Building cost I^C	260 USD per tonne	Replacement cost of an EAF steelmaking shop
Capacity shares κ_{add}, κ_{conv}	0.70, 0.46	Adoptions observed
Time to build T^{add}, T^{conv}	1, 2 years	Steubenville; Lorain, Fairfield
Obligations D^C, D^X	$\bar{\lambda}$ = 206 USD per tonne of BOF capacity; levels L_f	Owners' pension and retiree health obligations
Reline C^R; aging C^A	17 USD per tonne; 2 a year beyond $\bar{v} = 15$ years	Three reline projects; assumption
Idling C^I, C^H; restart C^S	10 USD per tonne, 8 a year; 20	AK Steel Ashland; U.S. Steel Granite City
Furnace replacement C^F	99 USD per tonne on 97% of mill capacity	Replacement cost of an EAF; AK Steel Butler
Discount factor β	0.85	Appendix D.4.1

Notes: Inputs set outside the likelihood; values in USD of 2005 per tonne. U_t^r: national utilization of route r; U_i^{p}: plant i's utilization relative to its route. κ_{add}, κ_{conv}: EAF capacity of an add-on and a conversion, as shares of site capacity. D^C, D^X: obligations due on shutting the blast furnaces and on closure, $\bar{\lambda} L_f$ per tonne of BOF capacity; $\bar{\lambda}$: mean unfunded obligation per tonne at $L_f = 1$; L_f: the owner's level, 0 below 5 USD of unfunded obligations per tonne, 1 at or above 50, 0.5 between. C^A: added cost per year a furnace campaign runs beyond $\bar{v}$ years. AIST: Association for Iron & Steel Technology. EAF: electric arc furnace; BOF: basic oxygen furnace.

Output, product mix and cleanup. The plant utilization factor U_i^{p} of (1) is the plant's average output relative to its route's in the years before the decision, leaving out the run-down years before a closure (Appendix A). The term $p_{it}^{\mathrm{mix},r}$ prices the plant's own product mix: with shares w_{ijt} of products j, prices p_{jt} and finishing costs C_{jt}^{fin} over the base product (hot-rolled sheet or rebar, with price p_{yt}^{base}),

$$p_{it}^{\mathrm{mix},r} = \sum_j w_{ijt} \left(p_{jt} - C_{jt}^{\mathrm{fin}} \right) - p_{yt}^{\mathrm{base}}. \tag{19}$$

The high-end cap $\bar{w}_t$, the largest share of cold-rolled and coated products in the EAF output of a works that adopted the EAF, applies to add-on hybrids, converted works and the EAF mills left when a hybrid retires its blast furnaces. The cleanup accrual C_{it}^{rem} per tonne of BOF capacity comes from the owners' annual reports (Appendix A) and is a lower bound, since firms book only what is probable and estimable.

D.2 Relines, idling and furnace replacements

The three hazards are fitted outside the likelihood; their estimation error is not carried into the intervals. The reline hazard $h^R(v)$ is a logit of the reline indicator on a constant, the campaign age, an indicator for the age cap and one for the years after 2000, fitted to 33 major relines in 554 works-years; it rises from 2.4 % a year at age zero to 23 % at age 24. The indicator for the years after 2000 absorbs the thinning of King's reline table and is set to zero. The idle hazard h^I is the share of operating site-years in which a site was idled, 0.33 % (9 in 2,733). The furnace-replacement hazard h_t^F is the rate of source-dated

replacements at EAF mills, 2.51, 1.24 and 0.71 per 100 mill-years in 1986–95, 1996–2005 and 2006–15.

D.3 Solution and likelihood

An idle site has the value $V_t(n,z) = -C^H K + \bar{v}_t(\mathcal{A}^{\text{idle}}(n) \mid n,z)$. A construction year pays the holding cost and the building-cost installment and moves to the next node without a choice. Within a menu $\mathcal{A}$ the probability that the owner chooses a is the logit

$$\Pr_t(a \mid \mathcal{A}, n, z) = \frac{\exp\big(v_t(a \mid n,z)/\sigma_i\big)}{\sum_{a' \in \mathcal{A}} \exp\big(v_t(a' \mid n,z)/\sigma_i\big)}. \tag{20}$$

For each spell and parameter vector, the stationary continuation $V_\infty(n,z)$ with the primitives of 2015 frozen solves (7) by value iteration, a contraction of modulus β. Backward induction then runs from 2015 to the spell's first decision year, with the expectation $\sum_{z'} \Pr(z' \mid z) V_{t+1}(n', z')$ over the price-state chain.

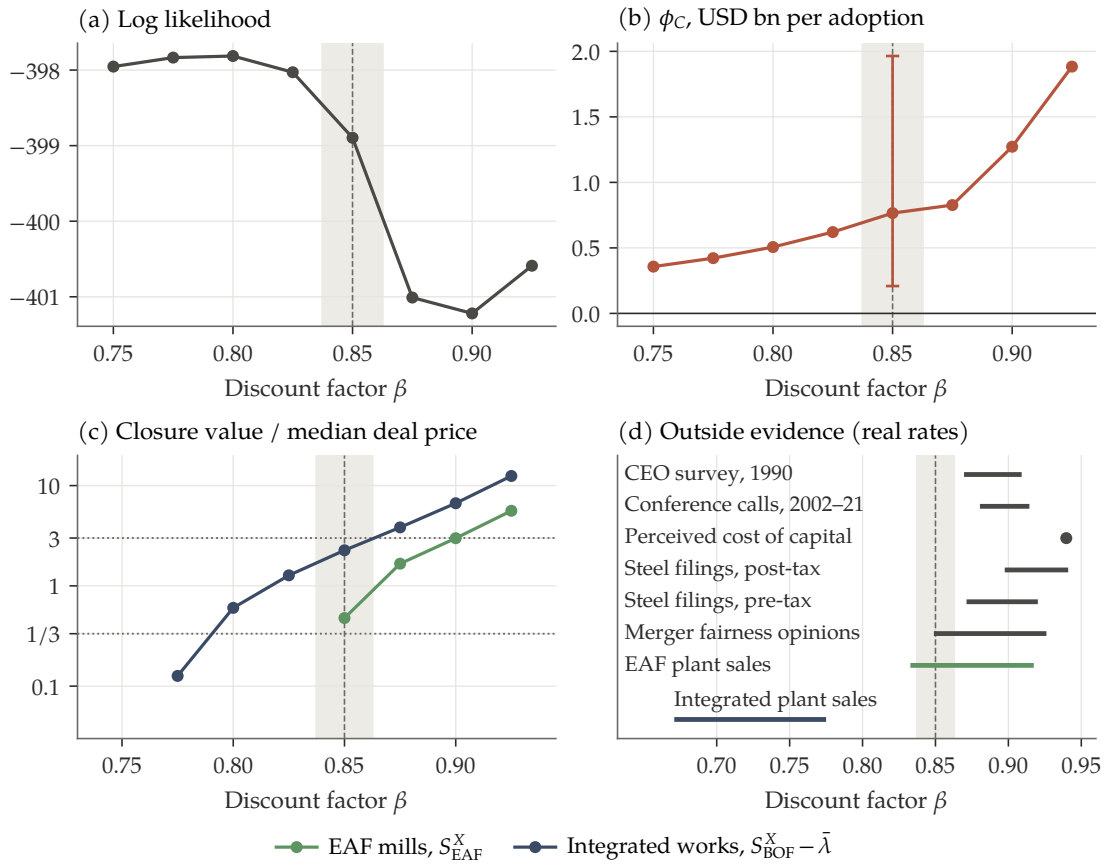
Each decision contributes the probability of the observed choice. Because relines and idling are events, that probability mixes the menus: keeping has probability $(1-h^I)(1-h^R)\Pr_t(\text{keep} \mid \mathcal{A}^{\text{reg}})$, a reline $(1-h^I)h^R \Pr_t(\text{reline} \mid \mathcal{A}^{\text{arr}})$, and a closure or conversion $(1-h^I)[(1-h^R)\Pr_t(a \mid \mathcal{A}^{\text{reg}}) + h^R \Pr_t(a \mid \mathcal{A}^{\text{arr}})]$; the choices of an idle site are logits within $\mathcal{A}^{\text{idle}}$. The log likelihood is maximized by L-BFGS on $(\phi_C, \log\sigma, S^X_{\text{EAF}}/100, S^X_{\text{BOF}}/100, \rho)$ with gradients from forward-mode automatic differentiation through the fixed point. A potential entrant's capacity follows the observed entrant sizes of its segment on a log-linear trend in the year of entry; its utilization factor is drawn from those of the observed entrants after it builds and is fixed thereafter. Given θ, the entry block (13) is concave in $(F^{\text{ent}}_{\text{flat}}, F^{\text{ent}}_{\text{long}}, \tau)$, which are concentrated out; the joint estimate holds ρ at its plant-model value.

D.4 Estimation and validation

D.4.1 The discount factor

I fixed a rule for β before estimating on a grid from 0.75 to 0.925: a value is plausible if both closure values lie within a factor of 3 of the median price buyers paid for going plants of that kind (Appendix C.3), S^X_{EAF} against 201 USD per tonne for EAF mills and $S^X_{\text{BOF}} - \bar{\lambda}$ against 119 for integrated works. Only $\beta = 0.85$ passes (Figure D.1). Below it the closure value of EAF mills is negative; above it that of integrated works net of obligations exceeds 3 times the median price. From $\beta = 0.825$ to 0.90, ϕ_C rises from 0.62 to 1.27 billion. The likelihood does not pin β down: its best value is 0.80, and the statistic of 0.85 against it is 2.2. The real rate that $\beta = 0.85$ implies, 18% a year, is high for a steel firm, and outside evidence points to a higher β (Figure D.1d): hurdle rates in a 1990 survey of chief executives (Poterba and Summers 1995) give 0.87 to 0.91, and conference calls in 2002–21 (Gormsen and Huber 2025), impairment tests, merger fairness opinions and sales of EAF mills give similar or higher values; only sales of

Figure D.1: The discount factor: likelihood, fixed adoption cost, closure values and outside evidence



Notes: The plant model re-estimated at each discount factor β; shaded: the value used. (b) Fixed adoption cost ϕ_C, with its 95% profile-likelihood interval at $\beta = 0.85$. (c) Closure values over the median deal price for plants of the kind, log scale: S^X_{EAF} for EAF mills; $S^X_{BOF} - \bar{\lambda}$ for integrated works, net of the obligation scale $\bar{\lambda}$; dotted lines: the selection rule's band, a factor of three either way. (d) $\beta = 1/(1+r)$ for real rates r; nominal rates are made real at the mean inflation of the GDP deflator over 2002–2020; bars span the range of each source. CEO survey: real hurdle rates reported by chief executives in 1990 (Poterba and Summers 1995). Conference calls: hurdle rates firms state in earnings calls, 2002–2021, and (dot) their perceived cost of capital (Gormsen and Huber 2025). Steel filings: discount rates in the impairment tests and valuations of steel firms' 10-K filings, 2008–2021, after tax or of unstated basis, and before tax. Merger fairness opinions: discount rates in advisers' cash-flow valuations in steel mergers of 2006–2008, from the proxy statements. Plant sales: the model's expected cash flow of the plants sold, over the price paid, in sales by sellers not in distress, 1985–2014 (Appendix C.3); green: EAF mills; dark: integrated works. EAF: electric arc furnace.

integrated works imply a lower β. I do not let it override the rule, because deal prices are a biased benchmark and a lower β can stand in for risks the model leaves out.

D.4.2 Recovery tests

The test re-estimates the baseline model on panels simulated with known parameters on the sample's own plants, states and price history, at two design points for the shock scale, $\sigma_0 \in \{90.5, 157.7\}$ USD per tonne: a low value at which an earlier version of the test passed, and the estimate. At each design point ϕ_C and the closure values are calibrated so that the simulated panels reproduce the observed numbers of closures and adoptions, which gives a true ϕ_C of 0.67 billion USD at the estimated scale. Coverage

uses all 100 replications per point; finiteness and width use the first 25. The criteria, fixed before any run and scaled by the larger of the true $|\phi_C|$ and the mean building cost of an adoption, are: (G1) the estimate is interior in at least 0.90 of replications; (G2) the coverage of the 95% likelihood-ratio interval lies between 0.90 and 0.99; (G3) the median bias is at most 0.25 times the scale; (G4) the interval is finite in at least 0.90 of replications, with median width at most 2 times the scale. The conditional test keeps the panels with exactly 2 simulated adoptions, as observed; this rule was set after the unconditional result of an earlier run was known.

At the estimated scale G1 to G3 hold, with coverage of 0.900, the interval is finite in 0.920 of the panels with a computed interval, since with no simulated adoption the likelihood keeps rising in ϕ_C, and its median width is 1.36 billion against a limit of 1.34. The test fails, unconditionally and conditionally, on G4 alone. At the lower design point (true $\phi_C = 0.02$) every criterion passes unconditionally in these 100 replications; the conditional test fails only in the conservative direction, with coverage above the upper limit of G2 and a median width above the limit of G4. The test supports the coverage of the interval and its lower end, not its precision.

The test of the joint estimator (Table D.2) draws entry counts from the entry model at entry costs calibrated to the observed entries. J3 to J5 pass at both design points, and J2 passes except for the entry-cost trend τ at the low one, where coverage is 1.00, above the upper limit. J1 fails: 88% and 76% of replications meet it. Every outer fit reports success; the failures are intermediate evaluations at which the gradient of the inner problem exceeds 10^{-8} or the Hessian is not negative definite, and the inner problem has converged at the final estimates in 100% of the 24 failing replications. Solving it more tightly there leaves the estimates of σ, S^X_{EAF}, S^X_{BOF} or the entry costs unchanged to three decimal places.

D.4.3 The profile and the firm bootstrap

The fixed adoption cost of 0.77 billion USD per adoption is 414 USD per tonne of the 1.85 million tonnes of new EAF capacity an adoption builds on average, and 222 per tonne of site capacity in Section 5.3. The 95% interval of a parameter ϑ is the set where the profile log likelihood $\max_{\theta_{-\vartheta}} \ell(\vartheta, \theta_{-\vartheta})$ is within $\chi^2_1(0.95)/2$ of the maximum. At $\phi_C = 0$ the model predicts 2.3 adoptions in 1985–2014 once σ is re-estimated (it falls to 91), against 1.6 at the estimate (Figure D.2); these counts come from histories simulated from 1985, while Section 5.2 sums the choice probabilities along the observed paths. Above the estimate the count stays close to the two observed adoptions, 2.0 at $\phi_C = 1.5$ billion, because a larger ϕ_C comes with a larger σ (225). What rejects $\phi_C = 0$ is the smaller shock scale it needs, which the closures and idling of EAF mills do not support.

The profile intervals treat the shocks of different plants as independent, and same-year closures of different sites of one firm co-move more than the model explains (permutation $p = 0.007$, but 0.57 without three firm-years; Appendix A.2). The firm bootstrap resamples firms with their plants, 100 draws, and re-estimates the five parameters, with the hazards and calibrated inputs fixed; all 100 draws converged, and the

Table D.2: Recovery test of the joint estimate

Criterion	Limit	$\sigma_0 = 157.69$	$\sigma_0 = 90.54$
Replications		100	50
J1 share converged	$\geq 95\%$	88% (fail)	76% (fail)
J2 coverage, σ	0.90 to 0.99	0.97	0.96
J2 coverage, $F^{\text{ent}}_{\text{flat}}$	0.90 to 0.99	0.93	0.98
J2 coverage, $F^{\text{ent}}_{\text{long}}$	0.90 to 0.99	0.95	0.94
J2 coverage, τ	0.90 to 0.99	0.97	1.00 (fail)
J3 |median bias| / truth, σ	$\leq 10\%$	0.1%	0.4%
J3 |median bias| / truth, $F^{\text{ent}}_{\text{flat}}$	$\leq 10\%$	1.0%	1.0%
J3 |median bias| / truth, $F^{\text{ent}}_{\text{long}}$	$\leq 10\%$	0.8%	0.6%
J3 |median bias| / truth, τ	$\leq 25\%$	0.9%	2.0%
J4 coverage, ϕ_C	0.90 to 0.99	0.94	0.96
J4 |median bias|, ϕ_C, USD bn	$\leq 0.25\times$ scale	0.057 (≤ 0.168)	0.034 (≤ 0.097)
J5 RMSE of σ, joint / plant model	≤ 1.05	0.99	0.96

Notes: Plant and entry decisions simulated with known parameters on the sample's plants, states and prices at two true choice-shock scales σ_0 (USD per tonne at median capacity), then re-estimated jointly. σ: choice-shock scale; $F^{\text{ent}}_{\text{flat}}$, $F^{\text{ent}}_{\text{long}}$: entry costs of flat- and long-product mills; τ: entry-cost trend; ϕ_C: fixed adoption cost. Criteria, set before the runs: J1, share of replications that converge; J2 and J4, share with a likelihood-ratio statistic at the true value of at most 3.841; J3 and J4, absolute median bias over the true value; for ϕ_C, absolute bias in billion USD, with its limit in parentheses, 0.25 times the larger of $|\phi_C|$ and the mean building cost of an adoption; J5, root mean squared error of σ, joint over plant model alone. RMSE: root mean squared error.

standard error of σ is 107 against the Wald 49. In 11 of the draws no adoption is drawn, ϕ_C is pinned only by closure and idling decisions, and the estimate lies between 5.8 and 28.1 billion; these draws put the 97.5th percentile at 12.3 billion, far above the profile upper end of 1.96. Among the 89 draws with an adoption the interval is 0.40 to 1.83 billion, inside the profile interval.

D.4.4 The gap decomposition

Let $\mathcal{N}$ be the seven parts of the adoption gap (11) (Table D.3), and $g_{it}(a, \mathcal{S})$ the gap computed with only the parts in $\mathcal{S} \subseteq \mathcal{N}$ switched on, the model solved again at the estimated parameters, so that $g_{it}(a, \mathcal{N}) = \Gamma_{it}(a)$ and the baseline is $b_{it}(a) \equiv g_{it}(a, \varnothing)$. The Shapley part of part k averages its effect over all orders in which the parts can be switched on:

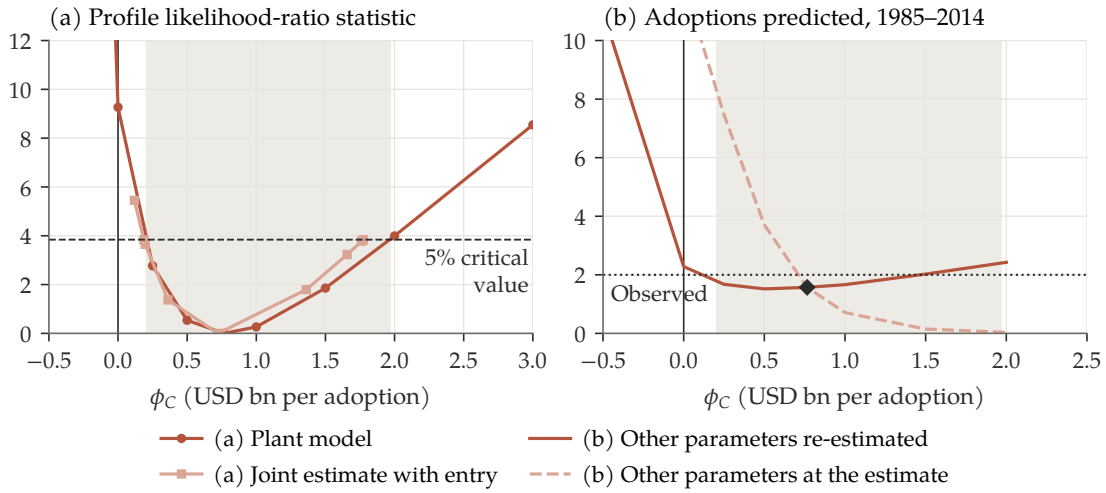
$$c^k_{it}(a) = \sum_{\mathcal{S} \subseteq \mathcal{N} \setminus \{k\}} \frac{|\mathcal{S}|!\,(|\mathcal{N}| - |\mathcal{S}| - 1)!}{|\mathcal{N}|!} \Big[g_{it}\big(a, \mathcal{S} \cup \{k\}\big) - g_{it}(a, \mathcal{S}) \Big], \tag{21}$$

and the parts add up exactly to the gap, (15). The baseline is not zero although a works and a converted EAF mill have the same flows and closure values, because a works draws a choice shock for each of its four choices and a mill for two. For the gap to convert it also holds site cleanup and the output lost while the new shop is built: 213 USD per tonne against 158, so the net gap to convert understates its cost by 54.

D.4.5 Re-estimates of the plant model

Table D.5 gives the re-estimates of Section 5.4, each changing one setting of the baseline.

Figure D.2: The profile likelihood of the fixed adoption cost and the adoptions it implies



Notes: (a) Profile likelihood-ratio statistic of the fixed adoption cost ϕ_C (billion USD per add-on or conversion), with every other parameter re-estimated at each value: plant model and joint estimate with entry; dashed line: the 5% critical value. (b) Adoptions in 1985–2014 predicted from 4,000 simulated histories at each ϕ_C, other parameters re-estimated (solid) or held at the estimate (dashed); diamond: the estimate of ϕ_C, where the two curves meet; dotted line: the 2 observed. Shaded: the 95% interval of the plant model.

D.4.6 Leave-one-out and fit

With one adoption left out the likelihood rises slowly in ϕ_C and the upper end weakens (Table D.5); leaving out Steubenville's later retirement of its blast furnaces gives 0.72 (95% interval 0.19 to 1.87). The model expects 5.9 flat-product entries in 1985–94 against 2 observed and 4.6 in 1995–2004 against 10: the linear trend of the entry cost cannot follow the wave of thin-slab mills of the 1990s. The decisions of 2015–2019, kept out of estimation, hold 3 closures and 1 conversion (Fairfield); for the decisions of 2016–2019 the predicted events are within about one expected event of the data (Table D.6), and Fairfield, an idle works, has a conversion probability of 0.0003 against a restart probability of 0.57. With so few events the holdout cannot reject the model.

D.4.7 The model without timing states

The plant model without idle state, relines or time to build, with the shock per tonne of capacity and the baseline's other inputs, gives $\phi_C = 0.47$ billion at $\beta = 0.85$, against 0.77 in the baseline. The comparison also changes the shock specification. The model of record has a smaller forgone profit (88 against 112 USD per tonne), a larger fixed adoption cost (222 against 137) and about the same obligation share of the net gap to convert (20% against 22%).

Table D.3: The parts of the adoption gap and how each is switched off

k	Part	Switched off
1	Building cost	$I^C = 0$
2	Forgone profit	Margin of the replacing EAF
3	Shutdown obligations	$\bar{\lambda} = 0$
4	Smaller plant	$\kappa_{\text{add}} = \kappa_{\text{conv}} = 1$
5	Product mix	No cap $\bar{w}_t$
6	Fixed adoption cost	$\phi_C = 0$
7	Closure value	Closing valued as EAF capacity

Notes: The seven parts k of the adoption gap and the input that switches each off; each part is its average contribution to the gap when switched on, over all orders. I^C: building cost per tonne of new EAF capacity; $\bar{\lambda}$: obligation scale per tonne of BOF capacity; κ_{add}, κ_{conv}: EAF capacity of an add-on and a conversion, as shares of site capacity; $\bar{w}_t$: cap on the new EAF's high-end share; ϕ_C: fixed adoption cost. Price of the technology: parts 1, 4 and 5; forgone profit: part 2; frictions: parts 3 and 6; part 7 is not part of adoption. EAF: electric arc furnace; BOF: basic oxygen furnace.

E Counterfactuals and the Remaining Blast Furnaces

E.1 The site-retention grant and the obligation takeover

The site-retention grant pays the old site to keep running clean capacity, where the open grant pays clean capacity wherever it is built. At 64 USD per tonne of EAF capacity a year it adds 8.8 adoptions and avoids 2.6 closures of integrated works, but it pays every tonne at an integrated site in every year, about a third of which would have run anyway: 33.8 billion USD, 109 USD per tonne of CO_2, the most of the instruments in Table 4. If only the owners of existing works respond, the takeover buys no output-neutral abatement: taking over half or all of the obligations closes 2.9 and 6.5 more integrated works (7.0 without policy) and gross CO_2 falls by 553 Mt at the full takeover, but steel that leaves the modeled sites is assumed to be made elsewhere at the blast-furnace intensity, so only a change of route counts as avoided.

E.2 What drove entry without policy

New EAF mills did most of the decarbonizing (Section 3.1) with no climate policy. I hold each force at its 1985 level in turn and express its contribution as the carbon price at which the world without the force emits, over 1985–2015, as much CO_2 as the fitted path. The forces are the EAF route's technology (non-material cost, flat-product quality penalty, high-end share $\bar{w}$, utilization), its input prices (metallic charge, electricity), both together, and the integrated route (costs, with the obligations kept at the works as if the PBGC had not taken over the plans).

The EAF route's technology did the work of a carbon price of 34.1 USD per tonne of CO_2: held at 1985, it leaves a world that emits 288 Mt more than the fitted path. The forces interact, so the attribution is not a split. With only the input prices held the world emits 421 Mt *less*, because the scrap price rose after 2003; with both held it emits 220 Mt more, an equivalent carbon price of 21.3 USD per tonne, 353 Mt above the sum of the single-force worlds, because cheap inputs matter only once the technology has improved. I report the worlds (Table E.1) and no additive shares. The abatement target

Table D.4: The net gap: shares by interval end and period

				Share of the net gap (%)			
	ϕ_C	Γ	Net gap	Technology	Forgone profit	Obligations	ϕ_C
A. Baseline estimate, gap to the add-on							
Lower end	0.21	400	273	46	32	0	21
Estimate	**0.77**	**581**	**413**	**25**	**21**	**0**	**54**
Upper end	1.96	975	732	7	12	0	81
1985–94	0.77	635	461	21	28	0	50
1995–2004	0.77	556	389	24	21	0	55
2005–14	0.77	540	381	31	11	1	58
Gap to convert	0.77	882	657	29	17	20	34
B. Model without timing states, gap to the add-on							
$\beta = 0.825$	0.34	463	334	41	29	−1	31
$\beta = 0.85$	0.47	531	372	34	30	−1	37
$\beta = 0.90$	0.92	764	498	19	31	−2	52

Notes: Means over works-years of running integrated works with an observed decision, 1985–2014. ϕ_C: fixed adoption cost; Γ: adoption gap, the one-time payment that would make adopting as good to the owner as the better of keeping and closing; net gap: Γ less the choice-set effect and the closure value; ϕ_C in billion USD, Γ and the net gap in USD per tonne of site capacity. Shares: technology, the price of the technology (building cost, smaller shop and product cap); forgone profit; and the structural frictions, obligations and ϕ_C. Panel A: the baseline at the interval ends for ϕ_C and by period, and the gap to convert; other rows: the gap to the add-on. Panel B: the plant model without timing states, shock per tonne of capacity, at its estimate for each β (Appendix D.4.7).

of 311 Mt is what the joint price buys on the fitted path.

E.3 Sensitivity of the ranking

Entry cost and elasticities. Table E.2 reruns the grants at the low and high demand elasticities of Appendix C.5, at the OLS and upper instrumented import-supply elasticities, and with the flat and long entry costs together at the ends of their 95% profile intervals (570 to 1204 USD per tonne for a flat mill). The elasticities move the open grant's budget by a few dollars and the adoption grant's by less; at the low demand elasticity the open grant is the cheaper. The entry cost moves the open grant through its base: at the upper ends of both intervals it costs 27.5 against 29.2 USD per tonne for the adoption grant, and at the lower ends it is far dearer. These are scenarios with both costs low or high, not a joint confidence region.

What an add-on replaces. Recounting the CO_2 of these works with choices and prices as estimated, but with the EAF also melting liquid iron from the blast furnaces, the adoption grant's small advantage reverses once that iron is 10 percent of the EAF's charge. *Displacement.* Under the open grant the part of the entrants' extra output that replaces other domestic output, $1 - \Delta X_y / \Delta X_y^{\text{ent}}$ for segment y over 1985–2015, is 0.87 in flat and 0.76 in long products; the rest goes to extra consumption and fewer imports. The open grant lowers average steel prices over 1985–2015, while the carbon price raises them (Figure E.1).

Table D.5: Re-estimates of the plant model

	ϕ_C (USD bn)	σ	S^X_{EAF}	S^X_{BOF}	LR	Log-lik.
Baseline estimate	0.77 [0.21, 1.96]	158	96	475	9.3	−398.9
EAF furnace replacement						
Without the furnace-replacement cost	0.77 [0.21, 1.97]	157	103	475	9.3	−399.0
Closure value						
S^X_{EAF} fixed at the median EAF deal price	0.51	108	201	480	–	−400.7
Capacity						
EAF capacity from the printed-capacity rule	0.77 [0.22, 1.93]	182	131	477	10.0	−400.0
Shutdown costs						
Gross obligation basis	1.21 [0.41, 2.90]	207	38	664	14.4	−398.6
Closure obligation lifted in Chapter 11	0.99 [0.32, 2.38]	178	68	429	12.6	−401.1
Obligation scale estimated (λ in place of $\bar{\lambda}$)	0.93 [0.24, 2.93]	176	77	573	10.1	−398.2
Margins and output						
EAF charge priced from scrap and pig iron	1.25 [0.57, 2.66]	203	80	470	26.5	−401.2
Output responds to the margin by route	0.79 [0.22, 2.00]	159	99	473	10.0	−399.1
With the largest elasticities	0.85 [0.24, 2.10]	166	104	493	11.1	−399.1
Engineering finishing cost for plate	0.96 [0.32, 2.27]	190	59	477	13.1	−399.7
Engineering finishing cost for long products	1.35 [0.58, 2.94]	245	27	468	21.7	−402.2
Engineering finishing cost for both	1.43 [0.66, >3.00]	259	24	472	23.9	−402.3
EAF flat margin +40 USD per tonne	1.30 [0.61, 2.77]	207	54	471	27.5	−401.9
EAF flat margin +80 USD per tonne	2.02 [1.11, >3.00]	285	−35	460	70.0	−405.0
Leave one decision out						
Without Lorain's conversion	0.98 [0.27, 2.52]	159	94	476	11.1	−391.0
Without Steubenville's add-on	0.89 [0.24, 2.33]	164	82	477	10.4	−393.1
Without Steubenville's blast-furnace retirement	0.72 [0.19, 1.87]	154	100	477	8.5	−396.7
Without Farrell's EAF site (closure coded twice)	0.79 [0.22, 2.02]	163	80	475	9.6	−394.6

Notes: Re-estimates of the plant model ($\beta = 0.85$), each changing one setting of the baseline, with ρ (the elasticity of the shock scale to capacity) estimated in every run. ϕ_C: fixed adoption cost, billion USD, with its 95% profile-likelihood interval in brackets (">3.00": still inside the interval at the grid's end). σ: choice-shock scale; S^X_{EAF}, S^X_{BOF}: closure values; all three USD per tonne. LR: likelihood-ratio statistic of $\phi_C = 0$; the run with S^X_{EAF} fixed has no profile (–) and an LR against the baseline of 3.51 (p-value 0.06). Leave-one-out log likelihoods drop observations (eight for the Farrell site) and are not comparable with the baseline's. EAF: electric arc furnace.

Table D.6: Held-out decisions: observed and expected events

	Observed	Expected
Closures of integrated works and hybrids	0	0.84
Closures of EAF mills	2	2.32
Add-ons and conversions	1	0.29
Idlings	0	1.08
Restarts	3	3.67

Notes: Decisions of 2016–2019, kept out of estimation, with their outcomes in 2017–2020. Expected: the sum of the model's choice probabilities at the baseline estimate, with the hazards and other parameters fitted outside the likelihood held at their estimation values and the inputs extended to 2020. Relines are not coded after 2015 and are left out. EAF: electric arc furnace.

Table E.1: What drove the replacement: each force held at its 1985 level

		Held at its 1985 level			
	Fitted path	Technology	Input prices	Both	Integrated
Adoptions, 1985–2014	1.6	0.3	7.3	1.4	1.7
Closures of integrated works	7.0	1.7	12.0	3.2	8.0
Closures of EAF mills	27.8	23.6	22.6	20.2	27.3
Entries of EAF mills, change	0.0	+2.7	+20.5	+15.5	+4.5
Integrated capacity 2015, Mt	47.9	69.8	10.3	61.9	45.5
EAF share of capacity 2015, %	58.3	48.0	91.8	58.6	61.1
Flat price, mean change, USD/t	0.0	+47.1	−27.3	+24.8	+9.9
CO_2 1985–2015, Mt	3,230	3,518	2,808	3,449	2,967
Change from the fitted path	0	+288	−421	+220	−277
Equivalent policy					
Carbon price (USD/tCO_2)		34.1	0	21.3	0
Adoption grant (USD/t)		542	0	333	0
Open grant (USD/t)		267	0	154	0
Obligation takeover (%)		>100	0	>100	0

Notes: Model outcomes, 1985–2015, with one force held at its 1985 level and other inputs on their observed paths; entry, choices and prices respond. Technology: the EAF route's non-material cost, flat-product quality penalty, high-end cap and utilization; input prices: the EAF route's metallic charge and electricity; integrated: the integrated route's costs, with the obligations kept at the works as if the PBGC had not taken over the pension plans (against the fitted path under the calendar-year rule, 3,244 Mt). Both: technology and input prices held together. Equivalent policy: the level, interpolated, that brings CO_2 back to the fitted path; >: not reached at the largest level run; 0: no more CO_2 than the fitted path. EAF: electric arc furnace; PBGC: Pension Benefit Guaranty Corporation.

The scrap price. The main case holds the scrap price on its observed path, but many new EAF mills raise the demand for scrap. In the first sensitivity the scrap price is multiplied, permanently and foreseen, by the ratio of the 25th or 75th percentile of the 1985–2020 real scrap price to its median, in the world and its baseline. At the abatement target the adoption grant costs 23.4 and the open grant 36.4 USD per tonne of CO_2 on the 25th-percentile path, with 78 percent of the open grant's payments made for entry that would come anyway, and 43.8 and 45.1 on the 75th-percentile path; baseline entry is 61 and 7 mills against 24. In the second, the U.S. scrap price rises by a share η^{scrap} per percentage point of EAF share above the baseline (Appendix C.7). Table E.3 gives net welfare at fixed policy levels, not at the abatement target.

Grants with a carbon price. Added to carbon prices of 25 and 50 USD per tonne, each grant set to avoid a further 311 Mt, the adoption and open grants cost about the same, 25.7 against 24.5 USD per tonne at the lower price and 29.5 against 29.9 at the higher, while the open grant pays far more for investment that would occur under the carbon price alone (66 and 73 percent against 24 and 32; Table E.4).

Grid emission factors. The main results count EAF steel at a fixed 0.4 tonnes of CO_2 per tonne. In this check its intensity is 0.08 tonnes of direct emissions plus 0.52 MWh of electricity per tonne times the CO_2 factor of the power sector of the plant's state in each year, as in the decomposition of Section 3.1. Without a carbon price no choice depends on emissions, so at each grant level choices, payments and prices are those of the main

Table E.2: Budget per tonne at the abatement target across elasticities and entry costs

	Adoption grant		Open grant		Conditional grant		
Case	USD/t	Anyway	USD/t	Anyway	USD/t	Anyway	Gap
Central	27.5	13	28.1	54	33.4	21	+0.6
Demand elasticity, low	27.7	13	26.5	56	34.3	21	−1.2
Demand elasticity, high	27.1	14	30.7	52	31.5	22	+3.6
Import elasticity, low	27.5	13	28.4	54	33.0	21	+0.9
Import elasticity, high	27.4	13	27.8	54	33.2	21	+0.4
Entry cost F^{ent}, low	26.2	14	60.6	83	27.6	23	+34.4
Entry cost F^{ent}, high	29.2	13	27.5	17	42.1	19	−1.7
Where the gap reaches zero							
		Range of the input			Zero at, 311 Mt	Zero at, 150 Mt	
Demand elasticity, flat		0.05–0.40			0.15	0.07	
Import elasticity, flat		0.70–3.17			5.67	8.15	
Entry cost F^{ent}, flat, USD/t		570–1,204			–	–	

Notes: Public spending per tonne of CO_2 avoided (USD/t, undiscounted), each grant at the level that avoids 311 Mt. Anyway: percent of the payments made for investment that would happen without the grant. Gap: open less adoption grant. Cases: demand and import-supply elasticities at low and high values; both entry costs F^{ent} of new mills at the ends of their 95% intervals. Lower panel: the input value at which the gap is zero, interpolated through the cases (extrapolated outside the range), at the target and at 150 Mt; –: with the long-product entry cost at its estimate the gap does not reach zero within the tested interval (Figure 9).

runs, and only the count of CO_2 changes: at unchanged grant levels the CO_2 each grant avoids falls by 4 to 6 percent. With each grant interpolated to the same abatement target, the adoption grant costs 29.6 and the open grant 29.5 USD per tonne of CO_2, against 27.5 and 28.1 at the fixed intensity, and the shares paid for what happens anyway hardly change.

E.4 What the counterfactuals hold fixed

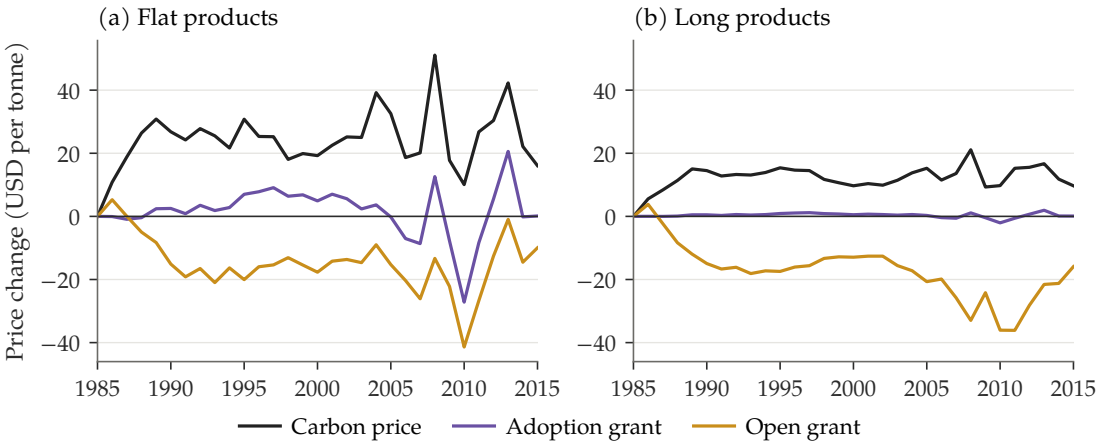
Table E.5 lists what responds, what is held and the likely direction of the bias. In the experiments, and in the entry costs estimated for them, each potential entrant is a mill of the mean size and utilization of the observed entrants of its segment; only the joint estimate draws them as in Section 4.3. The model has no bankruptcy state, so an owner in Chapter 11 carries its full obligation (Section 4.2).

E.5 Welfare and incidence

Net welfare is Equation (17) at the EPA value of 147 USD per tonne of CO_2, discounted at 2%, with $MCPF = 1.0$; Table 4 gives the break-even carbon values. The ranking depends on the choice-shock term in producer surplus. Over all 20 abatement targets, with the term the open grant leads at 19 and the conditional grant at 1; without it the takeover leads at 13 and the carbon price at 7. Without it the takeover does well because obligations are transfers for the economy but costs for the owner, so they keep open works whose closure would raise welfare.

At the abatement target the conditional and site-retention grants raise the surplus of existing plants by 18.4 and 16.8 billion USD (Table E.6). By region, the Midwest loses the most works jobs under every instrument, and the open grant's new mills add jobs in

Figure E.1: Steel price changes at the abatement target



Notes: Change in the domestic steel price against the model's path without policy, USD of 2005 per tonne, 1985–2015, point estimate. Each instrument is at the level that avoids 311 Mt of CO_2 over 1985–2015, as in Table 4. Flat products: sheet and plate; long products: bars, rods and sections.

the South; counting the jobs at new mills, the three instruments lose a similar number of jobs on net (Table E.7). Without the choice-shock term the entrants' gain under the open grant turns into a loss (8.2 billion USD). The local cost is reported beside net welfare, not added to it: the earnings loss of displaced workers, 110,000 USD (2005) per works job lost in present value (range 81,000 to 204,000), at 496 works jobs per Mt of capacity closed, or 55 million USD per Mt. The loss per job follows Davis and von Wachter (2011): the present value of earnings lost after a layoff, as a multiple of earnings before it, times average pay in steel mills. For the capacity the open grant closes it is 0.8 billion USD, small beside the transfers. Appendix C.8 gives the county evidence on what a closure did to its local labor market.

E.6 Power as a second case

Electricity generation meets the two conditions of the results, a clean technology that is a different plant and an old plant with shutdown costs that fall due on closure: a gas plant or wind farm is not a retrofit of a coal boiler, and closing a coal plant's ash units with the ash left in place brings post-closure care for at least 30 years (EPA 2015). The decomposition of Section 3.1 on the generators of the Energy Information Administration (EIA 2025a; EIA 2025b), with plants as units and the intensity of each technology class fixed at its realized rate, shows the same pattern: capacity-weighted intensity fell from 0.591 to 0.424 tonnes of CO_2 per megawatt-hour between 2001 and 2020, a fall of 28%, of which entry of new plants accounts for 68%, changes at surviving plants, including conversions and new generating units, for 22% and closures for 13% (Figure E.2). Closures lowered intensity because the plants that closed were coal plants; in steel they raised it slightly on net, because clean EAF mills closed as well as old works (Figure 2). In power, sunk retirement costs delay retirement (Davis, Holladay, and Sims 2022), and rate-of-return regulated utilities keep legacy capital longer than a

Table E.3: Net welfare by scrap-price case at fixed policy levels

		Net welfare (USD bn)			
	Baseline entries	Adoption grant	Open grant	Carbon price	Both
Policy level		100	100	25	25, 100
Scrap price path					
Observed path	24.0	3.6	25.7	35.1	40.9
25th percentile	61.4	8.5	31.9	56.0	66.8
75th percentile	7.2	0.8	9.0	8.6	10.1
Feedback $\eta^{\text{scrap}} = 0.012$					
Mills only	24.0	0.6	10.2	12.2	14.0
With scrap suppliers		3.2	22.3	30.0	34.4
Feedback $\eta^{\text{scrap}} = 0.027$					
Mills only	24.0	−1.5	−0.1	−0.1	0.6
With scrap suppliers		2.9	19.7	26.4	30.4

Notes: Net welfare, billions of USD of 2005, CO_2 at the EPA's social cost of carbon, discounted to 1985 at 2%; each case against its own baseline. Policy level: grants in USD per tonne of new EAF capacity, the carbon price in USD per tonne of CO_2; both: the carbon price and the adoption grant together. Baseline entries: new mills without policy. Percentile paths: the observed scrap price times the ratio of the 25th or 75th percentile of the 1985–2020 real scrap price to its median. Feedback: the U.S. scrap price rises by a fraction η^{scrap} per percentage point of EAF share above the baseline; mills only: net welfare without the scrap suppliers' gain. EAF: electric arc furnace; EPA: U.S. Environmental Protection Agency.

cost minimizer would (Gowrisankaran, Langer, and Reguant 2026).

E.7 Inputs for the remaining blast furnaces

Projects and support. Governments pay for the direct-reduction EAF (DRI-EAF) route at old works and new mills alike: German aid of up to EUR 550 million for thyssenkrupp's Duisburg works (approved July 2023 (European Commission 2023)), the selection in 2024 of the Middletown works in Ohio for negotiations over a U.S. award of up to USD 500 million (Cleveland-Cliffs 2024; DOE 2024a) and Swedish aid of EUR 265 million (2024) for a new mill at Boden (European Commission 2024b). At old works it has not always been enough: ArcelorMittal dropped an approved project at its German works (EUR 1.3 billion in 2024) in June 2025 because it did not pay even with the support (European Commission 2024a; ArcelorMittal Germany 2025), and after Cleveland-Cliffs judged the Middletown project no longer made business sense, the U.S. Department of Energy (DOE 2026) announced in August 2026 a USD 500 million award to upgrade the works' blast furnace.

Cost inputs. The gap to DRI-EAF of Section 6.5 changes two inputs of the adoption gap, the building cost of the new shop and the cost of its metallic charge; everything else is that of the scrap EAF, and no parameter is re-estimated. Table E.8 gives the main inputs and their sources. The DRI plant adds to the steelmaking shop's 260 USD per tonne. The cost estimates of CRU (2024) give a low and a high building cost for a U.S. gas-based plant; Nucor's plant in Louisiana, the one reported U.S. project, cost 266 USD per tonne of steel capacity, between the two. The central case uses the high value, the other cases the low value and Nucor's cost. The charge is a little more than one tonne of

Table E.4: Grants added to a carbon price

	No carbon price		Carbon price 25		Carbon price 50	
	Adoption	Open	Adoption	Open	Adoption	Open
A. Abatement and response						
CO_2 avoided by the carbon price (Mt)	–	–	368	368	808	808
Further CO_2 avoided by the grant (Mt)	311	311	311	311	311	311
Level	351	137	296	104	304	106
Extra adoptions	+8.9	+1.9	+8.7	+1.7	+8.8	+1.3
Extra entries	+0.6	+18.3	−0.7	+14.9	−3.3	+14.6
B. Public spending (USD per tonne of CO_2)						
Budget per tonne	27.5	28.1	25.7	24.5	29.5	29.9
Share ι (%)	13	54	24	66	32	73
Added investment	23.8	12.8	19.6	8.3	20.1	8.1

Notes: Adoption and open grants on their own, as in Table 4, and added to a carbon price of 25 or 50 USD of 2005 per tonne of CO_2 on both routes. Added to a carbon price, each grant is set at the level that avoids a further 311 Mt of CO_2 over 1985–2015 beyond the carbon price alone, interpolated between the levels run; extra adoptions and entries are counted against the carbon price alone. Level: USD per tonne of new EAF capacity. Budget per tonne: grants paid over the further CO_2 avoided, undiscounted, without the carbon revenue. Share ι: the share of the payments made for adoptions and entries that occur without the grant; added investment: the budget times $(1 - \iota)$. The conversion grant was not run with a carbon price. The columns without a carbon price use the main comparison's finer grid of levels. EAF: electric arc furnace.

DRI per tonne of liquid steel at the DRI cash cost of the works and year, in place of scrap: pellets, natural gas at the works' state industrial price (EIA 2026a; EIA 2025c), electricity at the model's EAF price and other operating costs from the same cost estimates. At market pellets it cost about as much as scrap in 2015.

Transfer assumptions. U.S. Steel and ArcelorMittal USA operated their own pellet mines, so 10 of the 13 works are charged their pellets at the mines' cost, which lowers the gap by 84 to 85 USD per tonne, to 641 in the central case. The cost is Cleveland-Cliffs' reported rate for its U.S. mines, used for the other owners' mines (an assumption). The parameters were estimated on scrap-EAF adoptions and are used as they are, and the future is fixed at 2015, when the scrap price was the lowest of 2010–2015, so at the scrap prices of 2010–2014 the gap would be smaller. Hydrogen is not priced, and the variant without the cap on high-end products in Table 5 is an assumption, not evidence.

Table E.5: What the counterfactuals hold fixed

Item	Responds	Held fixed	Likely bias
Steel prices	Yes, along net demand	Foreign supply beyond the import elasticity	Open grant's gain overstated
Entry	Yes (segment mean mill)	Entrants' investment in finishing capability	Entry understated later
Closures, adoptions	Yes	Cannibalization across a firm's works	Not signed
Scrap price	Sensitivity	Otherwise observed path	Grants that add EAF mills overstated
Other metal charge	No: a multiple of scrap	Virgin-iron use as a choice	High-end EAF cost understated
Clean iron, electricity, ore, coke	No (observed paths)	The DRI-EAF route only in Section 6.5	Not signed
Downstream buyers	National demand curve per product	Switching costs, freight, regional proximity	Cost of instruments that act through closure understated; conservative for the adoption grant
Trade policy	Imports, net of leakage	No border carbon adjustment	Carbon price understated
Choice shocks in welfare	With and without		Open grant's lead rests on them

Notes: Likely bias: the direction in which holding the item fixed moves the results. DRI: direct-reduced iron. EAF: electric arc furnace.

Table E.6: Who gains and who pays at the abatement target

		Grants			
	Carbon price	Adoption	Open	Conditional	Retention
A. Change in surplus (billion USD)					
Steel buyers	−41.9	−3.4	30.5	−17.6	1.8
Existing plants	−22.5	3.5	−41.7	18.4	16.8
Entrants	13.9	1.7	24.5	8.4	−0.5
Government	49.2	−6.5	−7.1	−8.1	−23.5
CO_2 avoided at the EPA value	31.0	30.2	30.8	31.1	30.2
Net welfare	29.7	25.5	36.9	32.2	24.8
B. Without choice shocks (billion USD)					
Existing plants	−21.4	−10.6	−39.1	6.4	2.2
Entrants	2.6	0.2	−8.2	1.2	−0.1
C. Not included in net welfare					
Earnings of displaced workers (billion USD)	−0.4	0.2	−0.8	0.3	0.3
Integrated capacity closed (Mt)	10.8	−7.2	8.7	−8.5	−8.3
EAF capacity closed (Mt)	−1.5	2.8	12.7	−0.1	−0.1

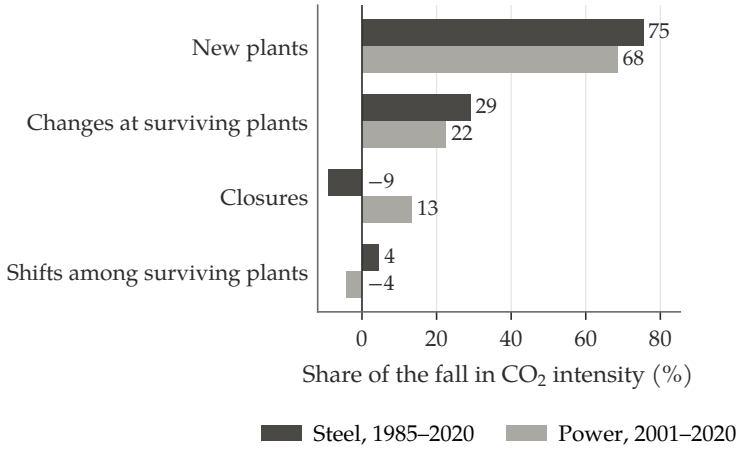
Notes: Changes against the model's path without policy over 1985–2015, each instrument at the level that avoids 311 Mt of CO_2, as in Table 4; billion USD of 2005, flows discounted to 1985 at 2%. Government: carbon revenue less grants paid, times the marginal cost of public funds, 1.0. CO_2 avoided valued at the EPA's 147 USD per tonne. Net welfare: the sum of the five rows above it, Equation (17). Panel B: surplus without the expected value of the logit choice shocks. Earnings of displaced workers: not added to net welfare; 55 million USD per Mt of capacity closed, negative for a loss, so an instrument that closes less capacity than the path without policy shows a gain (Table E.7 counts works jobs, including those at works that shrink or convert). Capacity closed: negative where an instrument keeps capacity open. The obligation takeover, which does not reach the target, and the conversion grant are not shown. EAF: electric arc furnace; EPA: U.S. Environmental Protection Agency.

Table E.7: Works jobs by region at the abatement target

	Carbon price			Adoption grant			Open grant		
	Lost	Gained	Net	Lost	Gained	Net	Lost	Gained	Net
Northeast	697	6	−691	303	0	−303	1,393	42	−1,351
Midwest	4,716	919	−3,797	2,719	86	−2,633	5,369	2,044	−3,326
South	1,054	1,685	631	1,129	159	−970	1,941	3,472	1,531
West	184	17	−167	117	1	−116	379	110	−269
Total	6,651	2,627	−4,024	4,268	247	−4,022	9,083	5,667	−3,415

Notes: Changes against the model's path without policy over 1985–2015, each instrument at the level that avoids 311 Mt of CO_2. Lost: works jobs at existing plants, including those lost when a works shrinks or converts; gained: jobs at new mills; net: gained less lost. Census regions; new mills are placed where earlier entrants of their segment located. Totals of jobs lost are those of Section 6.4.

Figure E.2: The fall in CO_2 intensity by channel: steel and power



Notes: Shares of the fall in the capacity-weighted CO_2 intensity of capacity, Equation (18), with plants as units. Steel: all plants in the panel, 1985–2020, the grid-based baseline of Figure 2; changes at surviving plants combine adoption, EAF steelmaking shops closed at surviving works and cleaner power (Appendix Table C.1); closures combine old works and EAF mills. Power: generators of the Energy Information Administration grouped into plants, 2001–2020, with the intensity of each technology class fixed at its realized rate. Shares add to 100 within each sector. EAF: electric arc furnace.

Table E.8: Inputs of the gap to DRI-EAF

Input	Unit	Value	Source
Building cost			
EAF steelmaking shop	USD/t steel	260	U.S. replacement cost (Section 4.2)
DRI plant, Nucor	USD/t steel	266	Nucor: 750 m USD for 2.5 Mt DRI (2014)
Prices in 2015			
Captive pellets	USD/t	65	Cleveland-Cliffs 10-K; assumed for others
Natural gas	USD/MMBtu	3.98–8.19	EIA, state industrial prices; range over works
DRI, market pellets	USD/t DRI	194–228	Recipe of CRU (2024); range over works
DRI, captive pellets	USD/t DRI	148–182	As above; 10 of 13 works, own mines (USGS)

Notes: USD of 2005, deflated with the GDP price index of the Bureau of Economic Analysis (BEA); t: metric tonne. DRI: direct-reduced iron, made from iron-ore pellets with natural gas. EIA: Energy Information Administration; USGS: U.S. Geological Survey; MMBtu: million British thermal units. Captive pellets: charged to the works whose owners, U.S. Steel and ArcelorMittal USA, mined their own ore, at Cleveland-Cliffs' reported cost of its U.S. mines including depreciation. Electricity is priced at the model's EAF electricity price. Building cost per tonne of new steel capacity: DRI plant cost per tonne of DRI capacity times the DRI per tonne of liquid steel. Prices of 2015 (pellets and gas nominal), for the 13 integrated works of Section 6.5. EAF: electric arc furnace.